



A Review of MAC Layer for Wireless Body Area Network

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Abstract

The Wireless Body Area Network (WBAN), which comprises of a set of tiny, invasive/non-invasive, light-weight, high-energy-efficient biosensors that monitor human health for early diagnosis and therapy, has lately received a lot of interest from researchers. As present and rising tendencies in communications included with the tendencies in microelectronics and embedded system technologies the existing techniques of IEEE 802.15.4 and IEEE 802.15.6 standards are explored in WBANs. As a consequence, discussed the challenges of MAC layer in WBAN. Secondly, different multiple access techniques along with TDMA, CSMA/CA, Slotted Aloha and Hybrid are explored in terms of design goals. In literature, a number of Medium Access Control (MAC) protocols for WBANs have been suggested for addressing the particular challenges associated with reliability, delay, collision and energy within the new research area. The design of MAC protocols is primarily based on multiple access techniques. Finally the general parameters of some popular MAC protocols are highlighted as performance metrics of WBANs.

Keywords WBAN · MAC · Scheduling · MA schemes · IEEE 802.15.6 · IEEE 802.15.4 · Castalia tool · TMAC · Tunable MAC · ZigBee MAC · Base- LineMac

1 Introduction

A Body Area Network is formally defined by IEEE 802.15 as, “a communication standard optimized for low power devices and operation on, in or around the human body (but not limited to humans) to serve a variety of applications including medical, consumer electronics/personal entertainment and other” [1]. T.G. Zimmerman in 1996 published an article and presented wireless personal area network [1]. Later on, the name WBAN evolved in 2001 by Van Dam instead after cable replacement for up to 10 meters such as Bluetooth, ZigBee etc. WBAN is a subset of Wireless Sensor Network (WSN) that has some similarities but many differences [2]. Table 1 shows a thorough comparison and challenges faced by WBAN. WBANs and WSNs are different from each other in terms of requirements and challenges and specific needs connected with the network-body connection are not addressed by WSNs [3].

As a consequence, the available frequency bands for WBAN are controlled by communication agencies in several nations. The brief description of frequency bands is shown in Table 2. Extensive WSN applications have been proposed to utilize shared spectrum of Industrial, Scientific, and Medical (ISM) bands excluding telecommunications [4]. The IEEE 802.15.4 standard supports on ISM band at 2.4GHz (2360MHz). These applications include health care monitoring, environmental sensing, forest fire detection, natural disaster prevention, and battlefield surveillance. The Medical Implant Communication Service (MICS) is valid in many western countries. It can't handle a high data rates. Wireless Medical Telemetry Service (WMTS) doesn't handle high-bandwidth applications as frequency range from 420 to 1429 MHz [5]. The Ultrawide Band (UWB) is defined in IEEE standard 802.15.6 and recommended for WBAN as operates on both high and low data rates. It has frequency range from 3100 to 10600 MHz. It operates with 3,8 channels and centered at 3996.3 MHz, 7987.2 MHz respectively. The Human Body Communications (HBC) is novel transmission technique for human body and has two frequency bands with a bandwidth of 4MHz, centered at 16MHz to 27MHz. Both bands are valid in USA, Korea and Japan [6].

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Table 1 WSN vs WBAN

Challenge	WSN	WBAN
Area Coverage	m/km	cm/m
Deployment area	Environment	Human Body
Node Size	Small/Large	Small
Node Bio-Compatibility	Not needed	strictly needed
Number of nodes	Hundreds/ Thou-sands	Fewer/Dozens
Node Task	Fixed or single task	Variable or multiple tasks
Homogeneity of nodes	Homogeneous	Heterogeneous
Network Topology	Static	Dynamic
Node replacement	Easy/Difficult	Difficult/Easy
Power supply	Accessible	Inaccessible
Node Life time	Months/Years	Days/months for on-body and months/years for in- body nodes
Data Rate	Homogeneous	Heterogeneous
Data Loss	Tolerable	Intolerable
Data Accuracy	Lower	Higher
Power Consumption	Higher	Lower
Data Accuracy	Lower	Higher
Data safety	Lower	Higher
IEEE Standard	802.15.4	802.15.6
Network Strength	Lower bandwidth connectivity	Higher bandwidth connectivity
Data Delay	Tolerable	Intolerable

Table 2 Frequency bands for WBAN

Frequency Band	Description
Medical Implant Communication Service(MICS)	It is licensed band specially used for implant communication and has range 402–405 MHZ. It is used in Japan, USA, Korea and Europe.
Wireless Medical Telemetry Service(WMTS)	This band is used for Telemetry Systems and has range 420–1429.5. It's popular in Japan and the United States, but it doesn't handle high-bandwidth applications.
Industrial, Scientific and Medical(ISM)	It is used in variety of applications and world-wide. It has a frequency range of 2400 MHZ and can handle large data rates.
Ultrawide Band(UWB)	It is defined in PHY layer of IEEE 802.15.6 standard and recommended for WBANs and has frequency range from 3100 to 10,600 MHZ. It is used in many countries.
HumanBody Communica- tions(HBC)	It is novel transmission technique for human body and has two frequency bands centred at 16MHZ to 27MHZ with bandwidth of 4MHZ. Both bands are valid in USA, Korea and Japan.

In field of health, Wireless Body Area Network has remarkably good scope in research and application expansion Via Biomedical Sensors (BMSs). These BMSs take physical parameters and transmit to wirelessly connected body area network coordinator (BANC) after converting into electrical signal [7]. BMSs are concise, smart and intelligent detect medically relevant parameters; Physical parameters like Blood Pressure, Temperature, Glucose, Electroencephalogram(EEG), Electrocardiogram(ECG), Respiration Rate, Oxygen Saturation, Physical Activity etc. Broadly Biomedical sensors are classified into three types : Physical Sensor (to measure body temperature, blood pressure,

blood flux, biological magnetic field, etc.), Chemical Sensor (detect the concentration of body liquid and ingredient of body liquid such as PH value, glucose, Ca+, etc) and biosensor (sense enzyme, antibody, DNA, antigen, microbe, hormone, DNA and RNA) [8].

Biomedical sensors can be categorized on the basis of type, application and placement. Further, BMS attributes can detect specific biological, chemical or physical quantity and converts it into a signal suitable for processing [9]. They provide high accuracy data, and the use of tissues, biological molecules and organisms has made biosensors unique for many diagnostic and real-time detection challenges, and

promises to continue this trend with their high potential for sensitivity and specificity. The BMSs are usually known in the world of WBAN as sensor nodes. These sensing nodes can be connected with the human body in three ways to establish the WBAN network [10] as shown in Fig. 12: Off-body sensors are kept a few centimeters from the body of a patient, on-body sensors known as wearable sensors, such as a wristwatch that is simply connected to the body or sensors sewed to a T-shirt or jacket and in-body sensors known as implanted sensors like the pacemaker is implanted under the skin of the chest, and there is no need for open-heart surgery. Similar published studies in this field include [11, 12], which investigates various elements of WBAN research such as the physical and data link layer as well as a number of low-power radio technologies. The approaches to health care with vital major contribution of this article is not only to examine and evaluate current low power communication technology in WBAN, it will also take into account the requirements and problems for communications in different situations of these low-power technologies.

The rest of this paper is organized as follows: Section 2 present motivation. Section 3 introduces the WBAN Three Tier architecture. Section 4 reveals the Mac superframe structure of IEEE 802.15.4 IEEE 802.15.6 and definitions of their phases. Section 5 presents the challenges of WBAN MAC Layer. Section 6, Multiple Access (MA) schemes are categorized. Section 7 through the analysis and summary of the vast literature, the MAC protocols of WBAN can be classified as— Scheduled-based MAC, Contention-based MAC and Hybrid-based MAC protocol. Section 8 represents performance metric of Castalia tool for simulation of WBAN MAC protocols. Section 9 concludes this paper. An important difference from other reviews is that this article covers current and future challenges of MAC WBAN design in the prospect part, giving readers with references and fresh ideas (Tables 3, 4, 5, 6, 7 and 8).

2 Motivation

With the increase of Population ageing, increasing healthcare costs and an increase in chronic disease prevalence are causing investors in the healthcare sector to monitor mobile health management and patient tracking. The issues of old age health care are becoming gradually more significant, as additional and multiple health issues occur with age [13]. The World Health Organization (WHO) states that, by 2050 the aging population will grow to approximately 25% of the global population, and this figure can even hit 33% for developed countries. Further, the global real-time health monitoring equipment market is expected to exceed \$67,982.2 million by 2022. These striking figures have increased the burden on healthcare

system as individuals will require more consistent and regular health monitoring. Nearly all nations spend enormous quantities of funds (billions and trillions of dollars) on their citizens' health care, according to documents accessible on the WHO website. The Fig. 1 shows total health expenditure per capita in the world from 2000 to 2025. Therefore, the development of smart and automatic role and significance in the coming years is in great demand. To fulfill this desire in current era, "An automatic wireless health monitoring system" for patients is one of the advanced and innovative technologies. Remote and wireless administration of patient monitoring and drug supervision has become possible by Wireless Body Area Networks (WBANs). Efficient WBANs covers health monitoring and treating health issues in remote areas to minimize hospitalization time and cost [14]. This system is used to measure vital statistics like heart rate, EEG, calories burned, respiratory rate, pulse rate, glucose, ECG, blood pressure of the patient and provide global accessibility of patients data to practitioners, doctors and any all kind of users via internet. Many researcher's priority is on developing a new energy efficient WBAN architecture, will result in a high degree of responsiveness in the sensitive situation and energy reduction. Their objective is therefore to develop:

- Sensor node to coordinator node or coordinator node to internet server: Develop approaches based on body area sensor networks that would enable the network with effective channel allocation for routine data, sensitive data and on-demand data transfer.
- It is important to establish strategies that will allow for reliable communication among body parts and between the sensors and coordinator node. Since reliable communication in body area sensor networks would offer maximum throughput, the research will focus on reliable data transfer with very little collision and prevent retransmission of lost packets.
- Implement strategies that will allow for speedy access to sensitive data and on demand information for managing crises as quickly as possible. Here, the major focus is on providing frequent access to, and availability of, sensor information, which will assist to optimize the lifetime of the network and maximize the accuracy of the service.

Advancement in this research field will result in improved and efficient BAN network. To allow energy efficient and rapid emergency response, researchers will boost the efficiency of WBANs in the future by minimizing delay, collision, re-transmission of missing packets, overhearing, idle listening, and over emitting.

Table 3 Schedule based MAC protocols

Author/Year/MAC protocol	Data Classification	Data Reliability	Energy consumption	Data collision	Remarks
Li, Huaming et al. (2009) [67] HBDMAC Protocol	No	Medium	Medium	Low	In HBD-MAC, instead of periodic beacon synchronization inherent heartbeat rhythm information is used for synchronization. The biosensor broadcasts a threshold value if the blood circulation increases or decreases so each sensor must require extra heartbeat sensor and rhythm signal system to receive the heartbeat signal, which is a big challenge in terms of energy saving for the protocol implementation.
N.Timmons et al. (2009) [68] Med-MAC Protocol	No	Low	High	Low	To achieve QoS requirement in terms of energy efficiency this protocol uses adaptive TDMA which implements dynamic adjustment algorithm for time slot allocation depending upon the classification of sensor nodes. The sensor data is ignored due to body sensors categorization. As a consequence high delay, Low Data reliability and higher energy consumption. Moreover, for high rate data it is not performing well.
K.S.Kwak et al. (2010) [69] Ta-MAC protocol	Yes	Medium	High	Low	The TaMAC protocol is intended to provide energy efficiency to a data transfer model for emergency traffic. Due to emergency information, TaMAC does not find traffic bursts on more than one sensor nodes simultaneously and does not consider periodic traffic. The time required for wake-up response signal and beacon sending results in high delay when transmitting emergency traffic. Results showed advancements in power consumption and delay over 802.15.4, Wise-MAC and S-MAC.
M.Ameen et al. (2012) [70] PE-MA Protocol	Yes	Medium	High	Low	PE-MAC has categorized traffic as routine or normal and random or emergency information. The behavior of BSN and BC are different on the classification of data. The limitation is that in the predefined time slot, each BMS either having routine data or sensitive-based, waits and then transmits data which degrades the performance of the MAC protocol, that is not suitable.
Behnaz Pourmohseni et al. (2013) [71] REE-MAC protocol	No	High	Low	Low	The proposed protocol is based on dynamic TDMA scheme to enhance packet delivery rate, channel utilization and to reduce duty cycle and power consumption which are basic requirements in WBANs. Basic Time Slot field (BTS) is dedicated for transmitting control packets and Dedicated Time Slot field (DTS) is dedicated for data receiving and transmitting. The emergency and on-demand data transmission leads to delay due to overhead packets and re-synchronizations which are not suitable for life threatening situations.

Table 3 (continued)

Author/Year/MAC protocol	Data Classification	Data Reliability	Energy consumption	Data collision	Remarks
Elli Kartsakli et.al (2014) [72] CLNC-MAC protocol	No	Low	High	Low	A cloud-based RLNC-based MAC (CLNC-MAC) protocol has constructed a statistical model to strengthen data delivery, throughput, and energy efficiency. In a cooperative network environment, it uses Random Linear Network Coding (RLNC) to ensure that all BSN information is transmitted to the point of origin without any failures through the network. This is done by the intermediating cloud controller. Due to relays higher delay in data transmission may be possible which leads to threatening of patient's life.
Zhan et al. (2016) [22] GTS based MAC protocol	No	Low	High	Low	The implemented scheme supports time-critical applications in beacon-enabled networks by enabling TDMA-based access. In this network, BC uses fixed time slots (GTS) to share data with end devices. Adaptive GTS allocation scheme is expected to address the shortcomings of traditional standards, which will allow efficient use of GTS resources. The end devices can request short, medium and long GTS with duration T-short, T-medium and T-long respectively. The lifetime of network suffers and more energy is consumed in GTS Request Command frame and beacon frame communication. In addition, it lacks flexibility due to the FCFS GTS scheduling algorithm.

Table 4 Schedule based MAC Protocols

Author/Year/MAC protocol	Data Classification	Data Reliability	Energy consumption	Data collision	Remarks
C, ALHAN, ALI et al. (2016) [73] NPP-MAC Protocol	Yes	Medium	Low	Low	In this scheme the Body Coordinator act as master and each body node required a synchronization for data transmission. The traffic is classified into high and low priority packets. Allocation of slots is based on FCFS so it might be possible the the emergency data drops packets and it needs re-transmission. So it consumes energy and loss of packets in waiting of time slot, these parameters reducing the performance of MAC protocol.
Yuan et al. (2016) [74] TTR-MAC Protocol	Yes	Medium	High	Low	It is proposed in this article Two-Round Reservation MAC, or TTR-MAC. Traffic is categorized according to the priority of the user and the index of safety severity. To minimize energy consumption, for burst traffic two round reservation technique is introduced and one-round reservation for periodic traffic. However, if the body nodes do not use the dedicated reservation slots, the other nodes cannot use that time slots as a consequences, the efficiency of network becomes low.
Lihua Ruan, et al. (2018) [75] Periodic monitoring based MAC protocol	No	Medium	Low	Low	An optimized framework for highly efficient smartBAN is designed in the proposed protocol, which significantly reduces the average data transmission delay. There is no adaptive channel allocation for emergency and on-demand information, as no priority has been granted. The results shows better performance in terms of energy savings and channel efficiency for periodic traffic. This scheme does not consider and investigate busy, on demand and emergency traffic which is the major requirement of WBANs.
Tae Min Hwang et al. (2018) [37] TDMA(GTS) Based MAC protocol	No	Medium	Low	Low	A TDMA-based self organization scenario is designed to transmit data from wearable body sensors to master device based on synchronization signal. Periodic messages are used to divide the nodes into clusters. To avoid collision each cluster is identified with unique number managed by master device. The protocol consumes energy during synchronization, cluster creation and cluster release operations. There is delay in emergency data transmission and packet loss might be occur.
Sun et al.(2019) [76] HP-MAC IEEE 802.15.6	Yes	Medium	Low	Low	Greedy strategy is proposed that is based on time slot allocation to reduce time complexity to solve the problem. As the transmission time and length of each node is altered, it is possible to increase the overall usefulness of all nodes for data transmission over a defined time period. But data gathered by sensors are not categorized and no low and high threshold values calculated.

Table 4 (continued)

Author/Year/MAC protocol	Data Classification	Data Reliability	Energy consumption	Data collision	Remarks
Wang et al.(2019) [77] Priority Aware Price Based power control (PPPC) scheme	Yes	High	Medium	Low	For inter-WBAN interference reduction, the proposed model is based on Stackelberg and a Bayesian game. A Stackelberg game is developed, Interference prices are set by the coordinators, and the active sensors alter their transmitting power appropriately. To optimize the transmission data rate of sensors while controlling the cumulative interference experienced by coordinators.
Xie et al. (2020) [78] A novel nest- based scheduling method(NBWS)	Yes	High	Medium	Medium	In this paper, to avoid interference the sensors in inter-WBAN or intra-WBANs are divided into clusters to increase the network life time by reducing energy consumption. All groups were scheduled according to a Time Division for Multimodal Sensor (TDMS) group scheduling paradigm.
Solt et al. (2020) Energy Efficient Heartbeat-Based MAC Protocol (HB- MAC) [79]	Yes	High	Medium	Medium	A proposed HB-MAC protocol for WBANs to efficiently support event-driven and irregular traffic relies on heartbeat-based synchronization and is consistent with Body Coupled Communication, in this paper. It solves key issues in body area networks by serving devices with inequities in capacities, to keep reducing the power consumption.
Ziaet al. (2020) G-MAC protocol [80]	Yes	Medium	Medium	Low	Traffic heterogeneity may be handled more wisely by grouping nodes of the same priority. Using a back-triggering/controlled traffic process by the coordinator, emergency and non-emergency traffics are managed. Without any modification in the IEEE 802.15.6 superframe all the phases are used. Each phase is dynamically modified based on the type of the traffic and the occupancy of the buffer zone.

Table 5 Contention based MAC protocols

Author/Year/MAC protocol	Data classification	Data reliability	Energy consumption	Data collision	Remarks
Alam et al.(2012) [29] TAD-MAC IEEE 802.15.6	No	Medium	Low	NC	In this Protocol on the basis of traffic status register bank each node dynamically adapts its wake up interval by CSMA/CA approach. It saves the extra power consumed by over hearing, collision, idle listening and unnecessary beacon transmission. But not suitable for WBANs due to based on the preamble approach. The nodes having emergency data has no priority and no traffic classification has been considered.
Iffat Anjum et al. (2013) [57] PLA-MAC protocol	Yes	Low	Medium	Medium	In the proposed Protocol the data is categorized as Critical, reliable, delay and ordinary data with highest to lowest priority respectively to reduce energy consumption. However, protocol consumes higher sensor energy as there is no difference between threshold values for vital signs during the contention phase.
Ullah et al. (2013) [82] R-MAC protocol	Yes	Medium	High	High	To classify the data, R-MAC protocol is proposed based on Radio Frequency Identification (RFID) Using CSMA/CA access. Since, CFP slots are allocated by the body coordinator to those nodes that receive channel access during the CCAP period. The suggested protocol reduces MAC outcome in terms of higher delay and collision as each body node claims to access channel during alarming situations.
Zhou et al. (2014) [83] RIFD based FCMA- MAC protocol	Yes	Low	High	Low	Fuzzy Control Medium Access (FCMA) implements phases of acquisition, fuzzy logic control and implementation to handle regular and sensitive data. During the CAP phase, the fuzzy rules prescribe the Contention Window Size for priority information and data speed. The suggested method is ideal for sensitive data, but sensory data is delayed in allocating slots to sensors during the contention to access channel sensors.
Chowdhury et al. (2014) [84] Slotted Aloha based MAC protocol	Yes	Low	High	Low	In this paper, a prioritized slotted Aloha scheme together with Discrete Time Markov Chain is proposed for non-saturated networks with finite body sensors. It tried to address collisions by halving the chance of re-transmission, which is constrained to a minimum in the following alternative slots. This scheme does not, however, focus on the information of the client, and the energy utilization of BMSs is high during contention phase.

Table 5 (continued)

Author/Year/MAC protocol	Data classification	Data reliability	Energy consumption	Data collision	Remarks
Hu, Long et al. (2015) [85] QAML-MAC protocol	Yes	Low	High	High	To improve QoS for the emergency data, this paper proposes multiple level-based design. Priorities given to various emergency data collected by emergency nodes. Each node send request for the time slot to router and then router update its priority by allocating time slot to node. More energy is consumed in process of updating the time slots on priority basis. Further data delay and collision may be occur which is not suitable for WBANs.
Soumen Moulik et al.(2017) [86] AT-MAC protocol	Yes	Medium	High	Low	A priority aware adaptive MAC-frame payload tuning mechanism is implemented and proposes adaptively tuned MAC protocol (AT-MAC) for WBAN to maximize the probability of successful packet transmission or reliability of the related body nodes based on real-time situation. The results are evaluated on saturated networks and non-emergency data is ignored which degrades the overall performance of MAC framework.
Ullah et al. (2017) [38] LTA-MAC protocol	Yes	Medium	High	Low	The authors proposes a Low-delay Traffic-Aware Medium Access Control (LTA-MAC) protocol to enhance performance of the MAC protocol. In IEEE 802.15.4 standard, during CAP period collision, data loss, delay and higher energy consumption occurs. To overcome this problem, Contention Differentiated Adaptive Slot Allocation CSMA-CA (CDASA-CSMA/CA) for scheduling access scheme is proposed, where guard bands are deployed between channels to eliminate overlapping of the designated channels.

Table 6 Contention based MAC protocols

Author/Year/MAC protocol	Data Classification	Data Reliability	Energy consumption	Data collision	Remarks
Nannan Li et al.(2018) [87] EIMAC-MAC protocol	Yes	Medium	High	Medium	Researchers proposes an energy efficiency and low inter-ferece (EIMAC) multi-channel MAC protocol. Channel mapping mechanism to categorize the states of channel. Authors try to reduce the contention period and congestion in single channel and decreases the delay up to some extend with increase in number of nodes as compared to IEEE802.15.6. However, higher energy consumed when Packet loss and delay occurs for low priority nodes.
Ambigavathiet al. (2018) [88] TPCAT MAC protocol	Yes	Medium	High	Medium	In this paper, the TPCAT- MAC (Traffic Priority-based Channel Access Technique) protocol is suggested to use IEEE802.15.6 to minimize the critical data packet transmission delay and resolve conflicts between other priority nodes during the back-off phases. To avoid overlapping issues a random backoff technique is developed. The proposed protocol provides better performance in terms of delay, energy consumption and throughput by reducing channel access time for critical as well as non-critical nodes.
Masud et al. (2019) [89] eTA-MAC proto- col	Yes	Medium	Medium	High	In this article first the BMSNs prioritize the trafficclass after finding the boundaries of the backoff period. Then each BMS perform the TCP-CSMA/CA (Traffic- Class Prioritization-based slotted-Carrier Sense Multiple Access/Collision Avoidance (TCP-CSMA/CA)) along with this the emergency traffic is handled through ETCP- CSMA/ CA (Emergency Traffic Class Provisioning-based slotted-CSMA/ CA) scheme. The TCPCSMA / CA system allocates a prioritized range of backoff periods that are unique and minimized for each traffic class in all backoffs while accessing the CAP channel. The problem occurs, during high emergency traffic raises data collision that minimizes the accuracy of emergency monitoring.

Table 6 (continued)

Author/Year/MAC protocol	Data Classification	Data Reliability	Energy consumption	Data collision	Remarks
Nithya et al. (2020) [90] P&CC-MAC protocol	Yes	Medium	Medium	Medium	In this article to minimize the delay, collision and energy utilization authors proposed two phase protocol named P&CC-MAC protocol. In first phase: To reduce collision Use runtime statistics to prioritize sensors and allow channel access only when potential nodes are sending data. In the second phase: The contention window (CW) size is estimated using the length of the queue and hit rate that accurately mimics the current channel state. Due CAP waiting time may be long, which will effect the performance of WBANs.
Samal et al. (2021) [91] TPLBS (Traffic Prioritized Load Balanced Scheduling) Protocol	Yes	Low	Medium	Medium	The authors concentrate on the waiting period of data packets that are waiting in the queue before transmitting to the access point. On the basis of criticality the data packets categorised and divided into four queues. For on-time delivery of emergency traffic TPLBS algorithm is proposed that make sure critical data packets are properly buffered.

3 WBAN Architecture

WBANs' network architecture usually consists of three levels. Fig. 2 presents a general outline of three-tier architecture [15]. The first tier is the lowest level consisting of an arrangement of intelligent physiological or Biomedical sensors (BMSs) working inside the wireless network; the second tier comprises of the medical server (PDA, personal computer, smart phone empowered by Internet); and third tier is an arrangement of remote servers dedicated to medical care and services associated with it (family members, Caretaker, doctor hospital, emergency department). Each level is a complex subsystem that integrates to provide performance, efficiency, mobility, protection, and cost savings.

- (a) First Tier: It is the patient body with wearable sensors that form a wireless network around the body to monitor changes in physiological parameters and provide feedback in real time to maintain an optimal health condition [16]. Sensor nodes on the human body can be put on or implanted deep within the skin (sensors in the body). Typically, biomedical sensors consist of sensor chip, micro controller, memory, radio transceiver and power supply. One or more physiological signals can be sensed, sampled and processed by sensor nodes. In a

network, these sensor nodes may be connected together in various topologies, such as star, mesh or tree. However, the most widely recognized is the star topology where the body nodes are associated with a central node or body area network coordinator (BANC) [17]. The types of commercially available sensors include accelerometer, blood pressure level, Blood Glucose level, ECG, EEG, EMG, Pulse Oximetry, Humidity and temperature sensors [18]. Wireless sensor nodes should have minimal weight, low power operation, seamless WBAN integration, low form factor, and patient-specific calibration.

- (b) Second Tier: The personal server (PS) or Base station (BS) constitute the second tier. This tier takes the data from BANC. So it first gathers, then processes, and then forwards the patient's vital signs to the medical server. It may use any low range communication protocol like Zigbee or Bluetooth to collect data from BANC. To interface the medical services, the IP address of the medical network is configured [19]. The PS or BS connects to the third tier using 3G communication or other long range communication protocols like General Packet Radio Service (GPRS), Wireless Wide Area Network (WWAN).

Table 7 Hybrid based MAC protocols

Author/Year/MAC protocol/Superframe Structure	Data classification	Data reliability	Energy consumption	Data collision	Remarks
M. A. Huq et al.(2012) [92] MEB-MAC Protocol	No	High	High	Medium	The authors suggested Medical Emergency Body MAC, named as MEB-MAC to enhance the performance of WBAN. The EAP is eliminated and short frames are used for emergency transmission. That also increases the overall results. The BC takes care of all the emergency data demands. In addition, the protocol has higher power consumption, which shortens the life of the sensor. This is mainly due to the elimination of complicated EAP and MAC access time.
Nourche`ne Bradai et al.(2013) [93] P-MAC Protocol	Yes	Medium	Medium	Medium	The body coordinator initiates the synchronization by transmitting the beacons as the start of communications in P-MAC protocol. To improve energy efficiency, the P-MAC protocol enables sleep mode. The idea presented in this protocol is theoretical and the performance is not measured by any concrete results. The superframe's overall structure is quite complex.
Choi et al.(2014) [94] MFS-MAC Protocol	Yes	Medium	Medium	Medium	The authors proposed modified frame structure (MFS-MAC), which is proportional to a number of nodes in WBAN, to reduce energy consumption. The downside of this method is that only when there is a node with data to be sent, the coordinator wakes up. The research emphasis is on BC. So, the resulting performance is comparatively low and the access delay compared to other protocols is also high.
xia et al.(2014) [62] Ada-MAC Protocol	No	High	Medium	Medium	The proposed protocol implemented schedule-based TDMA scheme for time "triggered-protocol" and "contention based" CSMA/CA mechanism. Not only can it allow dynamic allocation of guaranteed time slots, but it can also provide differentiated services for different nodes based on data types. For different type of traffic prioritization it uses multiple queues like IEEE 802.11e standard which is not a practical approach for WBANs.
Ibarra et al. (2015) [96] HEH-BMAC IEEE 802.15.6	Yes	Medium	Medium	Low	This Protocol incorporates two different methods of medium control, one being polling (ID-polling) and the other being probabilistic. This access protocol is specially designed for harvesting of energy in WBANs. Specific priorities and methods of channel access are allocated to the nodes. It depends on the importance of the data packet and the type of power supply.
Shu et al. (2015) [95] I-MAC IEEE 802.15.6	Yes	Medium	Medium	Low	This Protocol is proposed to improve the energy consumption, channel utilization, minimize data transmission delay. In their protocol, time slots can be divided into two different categories. Waiting time and delay of data transmission increases for all the other nodes when the body node has longer length data to transmit in one time slot.

Table 7 (continued)

Author/Year/MAC protocol/Superframe Structure	Data classification	Data reliability	Energy consumption	Data collision	Remarks
Jeongseok Yu et al.(2016) [96] CoR-MAC IEEE 802.15.6	Yes	Medium	Medium	Low	The authors propose to WBANs a hybrid priority-based MAC protocol based on contention over reservation mechanism to reduce sensitive data transmission delay. For sensitive data time slots are dedicated and other nodes can also transmit data over dedicated time slots by using traffic adaptive approach. In regular data, by increasing the number of sensor nodes, the average delay is increased.
Shakir et al.(2016) [97] Priority Assisted CSMA/CA-MAC	Yes	Medium	Medium	Low	To minimize energy utilization a hybrid prioritization scheme is proposed. At physical layer and MAC layer body nodes are prioritized, depending on the transmitting power. Furthermore, body nodes having similar priorities are further prioritized based on transmitting energy. Advance node awareness in the context of high density and diverse nodes is a difficult task.

- (c) Third Tier: The third tier also known as the Medical Server (MS) receives personal server data from registered patients. These data are available over the Internet to medical staff from their hospital office or family members for information. Using these data, the current patient status can be observed. If the data deviates from its threshold value, immediate action will be taken [15].

The development of wireless communication and popularization and portable computing has promoted the continuous progress of WBAN [20].

4 MAC Superframe Structure

One of the most significant aspects of developing WBAN standards is balancing QoS requirements with node power restrictions [21]. In existing models Bluetooth and ZigBee technologies have been modified for healthcare applications. Although these wireless technologies have been thoroughly researched and tested, they are inefficient. Since they are aimed towards more flexible networks that are utilized for greater transmission ranges. As a result, they are less energy efficient than protocols expressly intended for a WBAN (e.g., 802.15.6). Two IEEE standards are often used in designing of the MAC Superframe, as shown in Fig. 3, IEEE 802.15.4 and IEEE 802.15.6 [22] – [23] respectively).

4.1 IEEE 802.15.4 Standard

In a wireless personal area network (WPAN), the 802.15.4 standard is intended for data communication devices that operate at low data rates, low power, and minimal complexity (WPAN) [24]. Communications are possible at distances of up to roughly 10 metres, with maximum data transfer rates of 250 kilobits per second. A number of academics and industry people have been interested in IEEE 802.15.4 since its original release in 2003 [25]. It distinguishes between two kinds of channel access: beacon-enabled & non-beacon enabled. The non-beacon enabled option is solely dependent on contention. It manages channel access using unslotted CSMA/CA. In this method of operation, there is no provision for service demand assurances. The beacon enabled mode, to manage time-sensitive data transfers, on the other hand, provides a contention-free Guaranteed Time Slot (GTS) approach. Beacon frames are periodically sent by the Personal Area Network (PAN) coordinator to identify its PAN and to synchronize the nodes associated with it [26]. Figure 4 depicts the structure of superframe of the IEEE 802.15.4 standard. The superframe contains active and in-active periods. As part of the active period, all BMSs are required to conduct contention to get access to the channel during the Contention Access Period (CAP). The Contention

Table 8 Hybrid based MAC Protocols

Author/Year/MAC protocol/Superframe Structure	Data classification	Data reliability	Energy consumption	Data collision	Remarks
Lin et al.(2017) [98] CAP-MAC IEEE 802.15.6	Yes	Medium	Medium	Low	The authors presented, the Channel Aware Probing-based MAC (CPMAC) protocol overcomes the shortcomings of the sleep wake-up scheduling mechanism used in traditional static sensor networks, which can be very ineffective on WBAN. This protocol considers channel fluctuation of body sensors and transmission is only possible when the channel is strong enough to deliver fast and reliable communication. But it could be relatively inefficient to ensure fast and reliable data transmissions.
Darghamet al.(2018) [99]DTDMA/DS CDMA protocol	Yes	Medium	Medium	Low	The author proposes a hybrid DTDMA/DS-SSMA protocol that combines the advantages of DTDMA and DS-SSMA to adapt to changes in BMS traffic and various data requirements. To avoid idle listening, collisions and overhearing, the DTDMA scheme is used for periodic traffic. On the other hand, DS-SSMA is used for emergency traffic where nodes can concurrently transmit their data across the network, resulting in low induced delays even in high-traffic environments.
Jaeho Lee et al. (2018) [100]Asym-MAC protocol	Yes	Medium	High	Medium	The authors proposed hybridized (103) aSymMAC-Protocol based on slotted aloha, CSMA/CA and TDMA to differently considered energy balance between sensor nodes and body coordinator and immediate transmission of critical data to the BC regardless of any time constraint. To lengthen the life of the node, the suggested system shifts the coordinator's unavoidable energy usage to the coordinator. The latency of periodic data transmission increases and the energy efficiency of body coordinator decreases due to enough load which reduces overall throughput and degrades the performance of WBANs.
Yang et al. (2018) [101] CSMA/CA-TDMA hybrid MAC protocol	Yes	Medium	High	Medium	Taking advantage of CSMA/CA with collision avoidance and TDMA schemes, the proposed protocol uses a hybrid scheme. The load transfers to main station or coordinator to improve energy efficiency of body nodes. If the protocol uses an innovative TDMA scheme to save BSN energy during BSN reception time. It needs to improve the scalability of the protocol because the system would require more and more body nodes.
Ait Zaoui et al. (2019) [102] I-MAC protocol	Yes	Medium	Medium	Low	In this paper, it incorporates a hybrid WBAN model with IEEE 802.15.4 and IEEE 802.15.6 protocols. IEEE 802.15.4 is much more effective for sensors that require a low data rate, whereas IEEE 802.15.6 is better for sensors that include a greater data rate. This provides the mechanism for sensor interaction between the hybrid and the algorithm for protocol switching based on data rate and lifetime metrics.

Table 8 (continued)

Author/Year/MAC protocol/Superframe Structure	Data classification	Data reliability	Energy consumption	Data collision	Remarks
Yuan et al. (2019) [103] A-MAC Protocol	Yes	Medium	Medium	Medium	This article offers an adaptive MAC (A-MAC) technique for WBAN based on IEEE802.15.6. The durations of the contention and non-contention access periods are adjusted dependent on the number of nodes producing each way to optimize. During the non-contention phase, temporal synchronization between both the coordinator and sensor nodes is required, which consumes some additional energy.
Monowaret al. (2020) [104] Th-MAC (Thermal aware duty-cycle) Protocol	Yes	High	Medium	Medium	It introduces a framework that uses both contention-based and contention-free channel access methods to accomplish QoS in implanted biomedical sensors. ThMAC facilitates the trustworthy and an emergency data management technique ensures the timely transmission of unexpected emergency data.
Kim et al. (2020) [39] IEEE 802.15.6 MAC on NS-3 simulator	No	Medium	Medium	Low	On the NS-3 simulator, a practical scenario was designed to demonstrate the deployment effects of 802.15.6 MAC by implementing channel access schemes such as GTS, CSMA/CA, and polling schemes, association, connection-oriented link distribution, channel hopping method, and a two-hop star topology. The main focus of this article was on dynamic resource distribution, inter-WBAN interruption prevention, and intra-WBAN routing by analyzing average delay, power usage, and throughput.
Olatinwo et al. (2021) [105] A Hybrid Multi-Class MAC Protocol	No	Medium	Medium	Medium	The contention phase (CP) and the transmission phase (TP) are the two fundamental processes in the proposed protocol (TP). In the CP, only biomedical devices with health packets to transmit compete with equivalent probability for transmitting opportunity using a slotted ALOHA scheme, and the active biomedical devices are assigned a transmission time-slot using a reservation-based time division multiple access (TDMA) scheme in the transmission process.

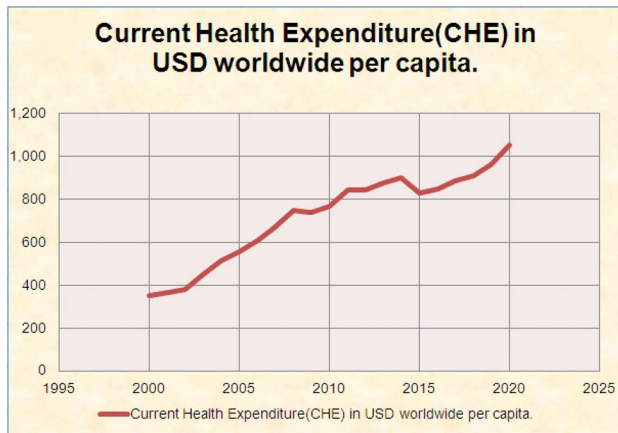


Fig. 1 Health expenditure per capita

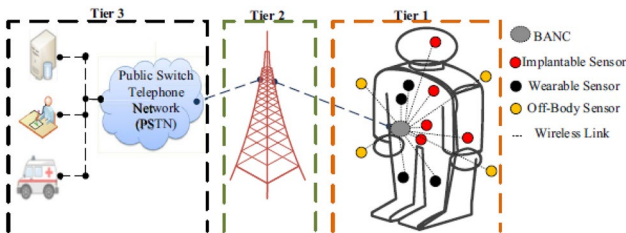
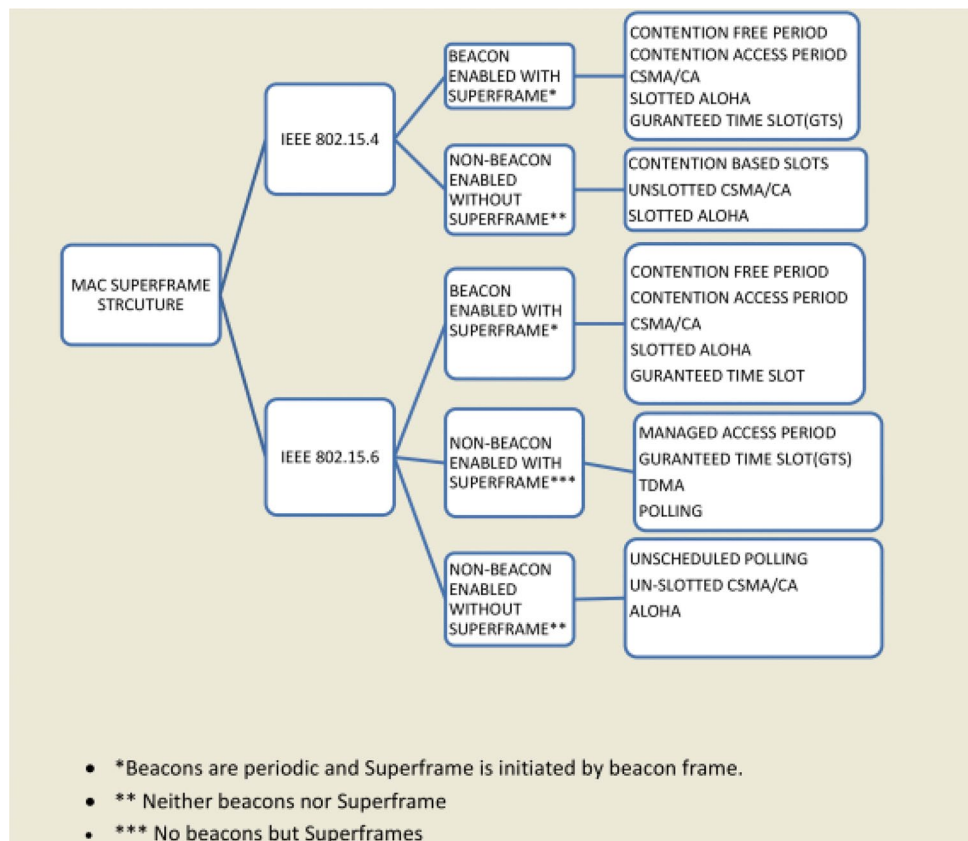


Fig. 2 WBAN: three tier architecture [15]

Fig. 3 MAC superframe structure



Free Period (CFP) is utilised for channel allocation only to those BMSs that have gained access during the CAP period [27]. BeaconInterval(BI) and SuperframeDuration(SD) are used by the PAN coordinator to design a structure for superframes. The time between two successive beacons is specified by BI, which include an active period and, optionally, an idle period. The active period, known as the superframe, corresponds to SD and is split into 16 equally-sized time slots during which frame transmissions are permitted. To save energy, all nodes may enter a low-power mode during inactive time (if it exists). Two factors, the BeaconOrder (BO) & the SuperframeOrder (SO), define BI and SD, respectively. BI and SD are defined in Eqs. (1) & (2) as follows:

$$BI = aBaseSuperframeDuration \times 2^{BO} \quad (1)$$

for $0 \leq BO \leq 14$

$$SD = aBaseSuperframeDuration \times 2^{SO} \quad (2)$$

for $0 \leq SO \leq BO \leq 14$

$$InactivePeriod = BI - SD \quad (3)$$

The superframe's minimal duration, is denoted by aBaseSuperframeDuration corresponding to $SO=0$. SD is further

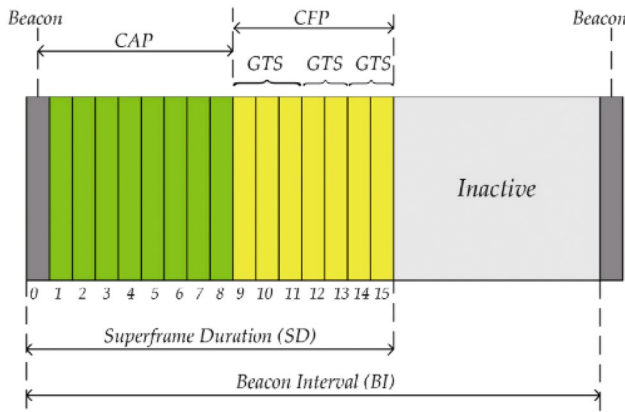


Fig. 4 IEEE 802.15.4 superframe structure [26]

divided into CAP and CFP. The coordinator sends the beacon at the start of slot 0 and the CAP follow shortly after the beacon. During the CAP, a slotted CSMA/CA algorithm is used for channel access. The communication in the CFP is done in TDMA (Time Division Multiple Access) mode, with a number of GTSs pre-assigned to each sensor for the use of devices that require dedicated bandwidth [26]. This standard is extensively utilized for WBAN test beds and simulations due of its benefits including energy efficiency, adaptability, and design flexibility. However, it is currently unable to satisfy all of WBAN's strict network criteria. The authors [28] have been discussed various strengths and challenges of IEEE 802.15.4 standard in the field of WBAN. As proposed in [29] for WBASN based on a traffic status register bank, the protocol dynamically adapts its wake-up interval. When traffic rates are stable or variable, the suggested approach allows the wake-up intervals to converge to a steady state, resulting in optimum energy usage. When a node wakes up, it transmits a wake-up beacon and then evaluates the next wake-up interval using the Traffic status register (TSR). Wake-up interval is updated for the time instant $t_i + 1$ based on the values of WUInt, and at time instant t_i as

$$WUInt(t_i + 1) = WUInt(t_i) + (\mu(t_i) + e(t_i)) \cdot t_{ref} \quad (4)$$

$WUInt$ = Wake-up-interval $WUInt(t_i)$ = Wake-up- interval at time instant t_i and $(\mu(t_i))$ is defined as

$$\mu(t_i) = \alpha \cdot X_1(t_i) + (1 - \alpha) \cdot X_2(t_i) \quad (5)$$

where $X_1(t_i)$ = least significant $X_2(t_i)$ = most significant and $\mu M(t_i)$ = Update factor and calculated on the basis of most and least significant halves computations along with constant weighting factor α as

$$X_1(t_i) = \frac{N_{0,1}}{L/2} \cdot N_{C_{0,1}} - \frac{N_{1,1}}{L/2} \cdot N_{C_{1,1}} \quad (6)$$

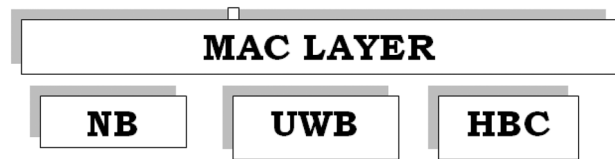


Fig. 5 IEEE 802.15.6 PHY AND MAC layers

$$X_1(t_i) = \frac{N_{0,2}}{L/2} \cdot N_{C_{0,2}} - \frac{N_{1,2}}{L/2} \cdot N_{C_{1,2}} \quad (7)$$

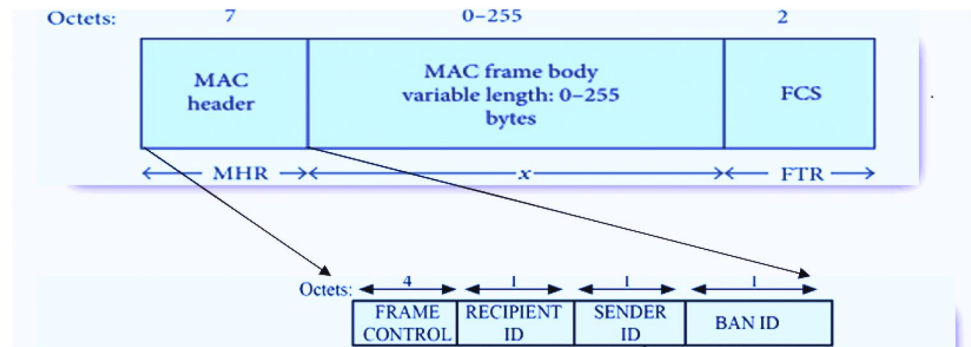
where $N_{0,1}$, $N_{1,1}$, $N_{C_{0,1}}$, and $N_{C_{1,1}}$ are the number of zeros, number of ones, occurrence of consecutive zeros, and occurrence of consecutive ones in X_1 , respectively. $N_{0,2}$, $N_{1,2}$, $N_{C_{0,2}}$ and $N_{C_{1,2}}$ are the number of zeros, number of ones, occurrence of consecutive zeros, and the occurrence of consecutive ones in X_2 respectively. As a result, it directs the modification of the wake-up interval towards the intended sequence. Finally, the previous wake-up interval value is added with and to compute $\mu(t_i), e(t_i)$ the next wake-up interval at the output. The TSR Length is modified as an optimization parameter to achieve rapid convergence, and its values for different traffic rate changes are provided. The GTS allocation is especially appealing for applications in time-sensitive WBASNs as it can increase data transmission reliability and performance due to planned transmissions. However, fixed first-come-first-served (FCFS) scheduling policies [30] may not well meet the time constraints of time-sensitive transactions, especially in medical and manufacturing sensor networks that monitor emergencies [30].

4.2 IEEE 802.15.6 Standard

This standard can be applied to a variety of medical and non-medical applications. It could be used to capture real-time health information for the detection and treatment of a wide range of maladies, such as cardiovascular disease, and it could also aid in the early detection of neurological issues. The WBAN must have a hub node, and a number of nodes as per network specifications. The nodes are organized into one or two WBAN stars according to the IEEE 802.15.6 standard [31]. In one hop star scenario, the every node is directly exchanging the data with hub node. On the other hand in two-hop star the node may exchange the data with hub node via relay-node. Which may sometimes delay the data delivery rate? This standard developed new PHY and MAC layers to address the requirements and problems of WBAN as illustrated in Fig. 5. The NB, UWB, and HBC are the three PHYs specified by this standard and their use is decided by the application.

On their top sophisticated MAC to control access to the channel. In Fig. 6 the IEEE 802.15.6 MAC frame format

Fig. 6 MAC frame format IEEE 802.15.6



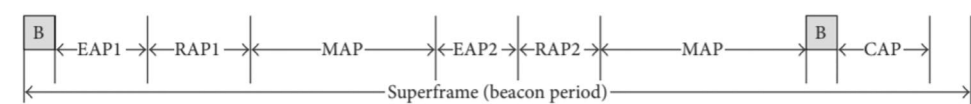
consisting MAC header of 56 bit, MAC frame body of variable length from 0 to 2040 bits and 16-bit frame check sequence(FCS).

The MAC header is further divided into four parts. The first 32-bits are Frame Control, second 8-bits are recipient ID, the third 8-bits are sender ID and the last and fourth 8-bits are BAN-ID. The frame control field contains control information such as the kind of frame, which might be a beacon, acknowledgment, or other frame control. The field of receiver and sender ID fields includes the recipient's and sender's address information correspondingly [32]. The WBAN ID includes WBAN information for the transfer to be active. The very first field of 8-bits in the MAC frame body consists freshness data needed for nonce generation and identification of replays. The payload field contains the data frame, and the last 32 bits of the message integrity code (MIC) provide information about the authenticity and completeness. The IEEE 802.15.6 supports the three communication modes [33] as follows.

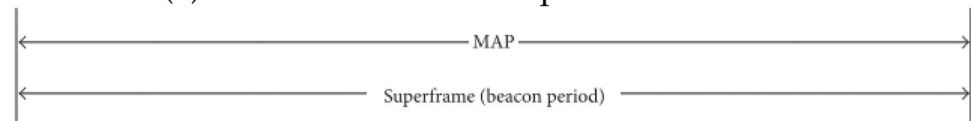
4.2.1 Beacon Mode with Superframe

The hub node is responsible for dividing the time axis into a series of superframes. Beacons of identical length surround these superframes. For data transport, each superframe has a certain slots that can be allocated for use. These slots are of the same length and are numbered from zero to 255. The hub sends out beacons to determine the superframe borders and assign slots. In non beacon modes, polling frames specify the superframe borders as beacons are not utilized. Except for idle superframes, the hub generally broadcasts beacon frames in every superframe. In addition, the hub node can change the offsets of the beacon times, causing the slots to move. When operating in beacon mode, the hub communicates superframe boundaries and synchronizes sensor nodes by regular beacon transmission as indicated Superframes are shown in Fig. 7a. In each superframe, a hub organises suitable access phases following beacons **B**. Beacons start a superframe, they are sent in each super frame's first slot. Their duties include identification of the coordinator,

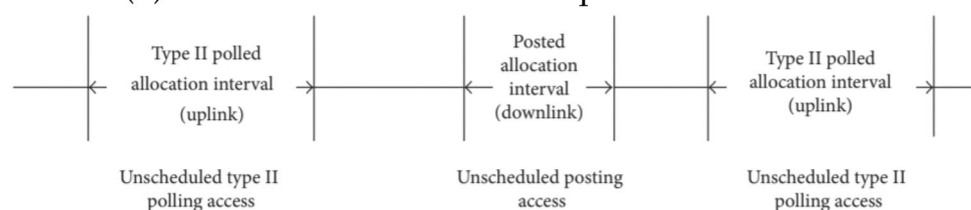
Fig. 7 Three communication modes of IEEE 802.15.6. **a** Beacon Mode with superframe structure. **b** Non Beacon Mode with superframe structure. **c** Non Beacon Mode without superframe structure



(a) Beacon Mode with superframe structure



(b) Non Beacon Mode with superframe structure



(c) Non Beacon Mode without superframe structure

network administration, power control, and synchronization of devices. The access phase of the hub can be placed in the sequence depicted in Fig. 7a: Exclusive access phases (EAP1 and EAP2) are for the transportation of high priority traffic or emergencies, Random Access phases (RAP1 and RAP2) are for regular transport where the data traffic is without the crucial time and events on request under their usual settings, Managed Access Phase (MAP) are for scheduling or poll nodes and may use TDMA to support these access modes and Content Access Phase (CAP) have been used for non recurring traffic and nodes use Slotted ALOHA or CSMA/CA methods. In the analysis model of the author [34], the number of competing nodes with priority levels 1 to 7 is limited. b_i & p_r be the probability that the channel is sensed busy (in an ideal channel condition) for a priority class i and the packet error probability respectively.

In Eq. (3) and (4) the P_i and P_r is calculated as

$$P_i = B_i + (1 - B_i)P_r \quad (8)$$

$$P_r = 1 - (1 - P_e)(H + E(P) + ACK) \quad (9)$$

where H and $E(P)$ represent header information or user data, ACK represents an acknowledgement packet, and the value of the bit error rate P_e is always equal to the average bit error rate (BER).

In Eq. (6) H is derived from PHY layer preamble represented as HP and header HPH

$$H = H_P + H_{PH} + M_H \quad (10)$$

on the other hand in paper (36) the P_r is derived as follows

$$P_r = 1 - (1 - P_e)^T \quad (11)$$

where T is the total length of payload (in bits), and derived as

$$T = \text{PREAMBLE} + \text{PHY H} + M_H + L + \text{RTS} + \text{CTS} + \text{ACK} \quad (12)$$

where PHY layer parameters are PREAMBLE and PHYH (Physical layer header). The MAC header is M_H and the actual information is represented by L . The control packets represents the transmission of RTS, CTS and ACK of packets.

As per study, it has been found that the b_i , τ_i and S_i are derived as follows in [36] and [34]

$$b_i = 1 - (1 - \tau_i)(N_i - 1)_{(j=0, j=1)}(1 - \tau_j)(n_j) \quad (13)$$

$$\tau_i = \frac{(\bar{X}_i)}{(\bar{Y}_i) + (\bar{X}_i)} \quad (14)$$

where $\bar{X}_i$ = average attempts by the tagged node in the priority class i , $\bar{Y}_i$ = average backoff time experienced by same node.

τ_i represents the transmission probability of a node in the priority class i . It is calculated as shown in equation [14]. S_i as depicted in equation [15] be the probability that exactly one node in the priority class i transmits on the channel. Under the premise of a finite nodes under stable and lossy channel circumstances, this work presents a reliable mathematical model to estimate the throughput, energy consumed, and latency of this protocol for various priority classes.

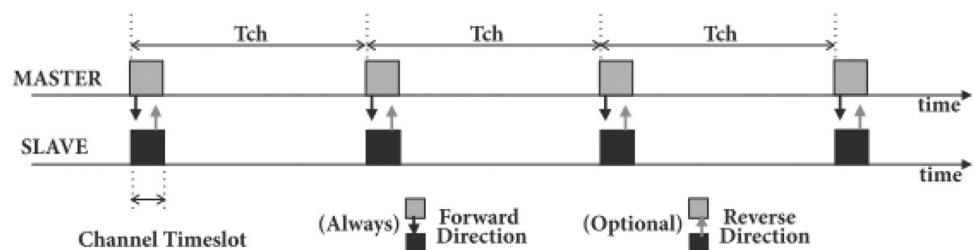
$$S_i = n_i \tau_i (1 - \tau_i)^{n_i - 1}_{(j=0, j=1)} (1 - \tau_j)(n_j) \quad (15)$$

4.2.2 Non Beacon Mode with Superframe

The Non-Beacon-enabled mode can offer better data throughput but at the cost of significant energy consumption, mostly as a result of the idle listening issue. As shown in Fig. 7b Managed Access Phase (MAP) are for scheduling or poll nodes and may use TDMA protocol to support these access modes. As [37] study suggests a way to create a self-organizing, TDMA body area network to provide multi-signal synchronization and to evaluate its correctness with the CSMA/CD technique. With TDMA's wireless communication technique, the suggested WBAN can achieve different critical signals, which are crucial for time synchronization with heart signal. As illustrated in the Fig. 8, a single channel is assigned to the TDMA slot.

There is a limit to the channels that may be linked at a time, dependent on the time slot size. When the length of one cycle time is T and the length of the time slot is t_{slot} , as the synchronization signal channel always occupies one slot the number $n_{channel}$ of the connectable channels can be expressed as follows:

Fig. 8 TDMA: Allocation of time slots [37]



$$n_{channel} = \frac{T}{t_{slot}} - 1 \quad (16)$$

where $n_{channel}$, t_{slot} and T represents number of channels, length of timeslot and Length of one cycle time respectively.

Fig. 9 depicts the process of transferring data in a planned slot based on a central device's synchronization signal. It is possible to gather data from all devices at the same time by obtaining data from each device in a single cycle without any collisions. Now, synchronization signal has the same cycle as transmission signal. In the case of TDMA, the synchronization signal is used to discriminate between the signals in a cycle and to prevent communication collisions in the system.

Polling is used during the MAP period to determine the planned allocation intervals for the uplink, downlink, and bilink services. Unscheduled bilink allocation intervals can likewise be handled using this method. As a result, data cannot be instantly sent to BMSs from the master node. Instead, the master node should first transmitting an active alert to the target BMS. Second, MAC without beacons can only collect one type of patient data at a time, which is inappropriate in life-threatening situations.

4.2.3 Non Beacon Mode Without Superframe

This structure of Superframes does not implement predetermined times to all the sorts of patient data are transmitted, however it is designed for Type-II scenarios [38]. Without any borders of superframes, every node develops its own time frame basis in the non-beacon mode without bounds. Type II unplanned bi-link appeal when your hub is operated in a non-beacon mode [6]. A hub delivers a packet to establish connection assignment to the node with Type-II unscheduled bi-link assignment to provide an unscheduled bi-link allocation requested by the node. The slot allocation to BMSs in this Superframes is based on contention or post-contention [39]. With an unplanned interval of allocation of two links, a pivot or T-poll frame is sent to the node to provide improvised uplink access, or a control framework is transferable to the node via a downlink. The benefit of this arrangement is that non-emergency-based BMSs do not disrupt the operation of emergency-based BMSs. The preset

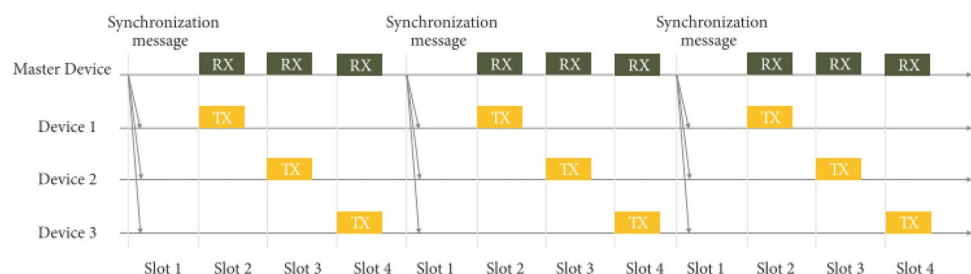
allocation of slots to one sort of sensory input, on the other hand, is a waste of slots. It is a waste of slots to limit the allocation of slots based on preset criteria to a single type of data. In nonbeacon enabled mode, a coordinator node does not transmit any beacon and nodes do not need to be synchronized. GTS is not permitted in this mode. All the transmissions use un-slotted CSMA/CA or slotted Aloha to access the channel. Non-beacon mode has the advantage that the node does not have to be powered up every time a beacon is received. Coordinator node must wait to be polled by each node before communicating with them. The third mode is typically destined for entertainment purposes such as electronic games, etc.

5 MAC Layer Challenges In WBANS

Real time and location independent health monitoring made possible through WBANs from the last decade. The network area of WBAN is human body so system devices must be small in size, light weight, low in complexity, power efficient and easy to use. For communication in WBAN various factors are important like data reliability, delay, throughput, energy consumption, life time, transmission time etc [40]. But during sensing, data processing and wireless communication sensor nodes consume lots of energy. Overall, these factors affect WBAN's service reliability and energy efficiency. Reducing the energy use of body nodes, addressing and channel access control mechanisms of MAC layer and network layer routing protocols play vital role. The issues like suitable channel allocation, idle listening, overhearing, congestion control, packet delay, re-transmission, data reliability, collision control and protocol overhead that fall in the scope of MAC layer [41]. Further, existing MAC protocols of WSN are not well suited to support WBANs due to constraints of limited resources on human body [42]. WBAN MAC protocols need to consider high data reliability, low packet delay, less collision rate, avoid re-transmission, overhearing, idle listening, over emitting, energy efficient and high throughput.

In the literature, several energy efficient, low powered, traffic adaptive, static, dynamic and hybrid MAC protocols have been discussed to find the gaps and limitations of

Fig. 9 Multidevice connection: Allocation of time slots [37]



existing protocols for WBAN. A brief overview of existing MAC Schemes and MAC protocols are discussed below. In this section try to find out various issues in existing MAC frameworks. Given its unique constraints and characteristics, designing and developing MAC protocols for WBANs is a challenging task [43]. Many MAC protocol techniques proposed by different researchers in the last couple of years. MAC layer issues and challenges that WBANs should address while proposing a MAC technique are discussed below.

5.1 Heterogeneous Nature of Data

Different varieties of BMSs are used to monitor various vital symptoms. Since different sensors have different frequency and data rates, they require different computing power, energy and storage requirements to transfer data to the body coordinator node [34]. The development of MAC protocols that meet the heterogeneous data rate requirements for BMSs is therefore important.

5.2 Waiting Period Based Slot Allocation

The slot allocation based on the waiting period is performed with a scheduled TDMA (Time-division multiple access) access scheme. It requires synchronization in the pre-allocated time slots before data transmission [44]. The nodes with sensitive data require immediate transmission of data without delay. Delay results in higher energy consumption and lower data reliability which is not suitable for critical information [45].

5.3 Contention Based Slot Allocation

Studies have noted that MAC schemes use contention based like CSMA/CA scheduling schemes in which all forms of nodes contest access channels during the contention access phase (CAP) time [46]. The routine data based nodes do not care about giving priority to sensitive data-based nodes in the allocation of slots. This results in energy consumption, more number of collisions and re-transmission of not acknowledged packets [47].

5.4 Channel Access Design Complexity

The division of the frequency band allocated to devices for communication. In FDMA (Frequency Division Multiple Access), the bandwidth is divided between different frequencies and allocated to the sensor devices. They transmit data on allocated frequency band. Collision-free data transmission is provided by the FDMA, but hardware design is a challenge [48].

5.5 Threshold Values Based Slot Allocation

Existing research does not agree to assign slots to more than one vital sign on a priority basis when two vital signs are seen concurrently with lower-to-higher or higher-to-lower threshold values. As stated in [12], in more life-critical situations, the lower threshold of a vital sign is essential than the high threshold of vital sign. In addition, the MAC superframe architecture has to add a separate slot for each type of sensitive data in order to solve this challenging problem [49].

5.6 Alert-based Slot Allocation

The allocated slot is utilized by the applicable BMS to transmit a warning signal and assign emergency slots. The node actually requires a three-way handshaking system to be setup a body coordinator interaction session and data transfer [50]. This method reduces the reliability of body node information due to the long waiting time for a spot to be allocated.

5.7 Permission-based Slot Allocation

The slot allocation on permission to body nodes consumes higher energy with higher delay [51]. When waiting for beacon to forward and receive data, these difficult problems are not ideal for sensitive body nodes.

5.8 Preemptive and Non Preemptive Based Slot Allocation

When sensitive data are received, the body coordinator precludes routine based data from allocated slots and assigns emergency data [52]. Blocking and deleting routine data from the assigned slots decreases protocol performance in terms of low data quality with a high fall in data from a client, and BMSs use more energy in re-transmitting the missing packets.

5.9 Hybrid Based Slot Allocation

Hybrid scheduling schemes, combination of access schemes like TDMA and CSMA/CA [50]. All scheduling systems provide an optimal solution for daily software allocation of slots. Contention allocation and pre-defined access-based slots to sensitive nodes. In terms of increased data collisions, higher retransmission of lost packets, and

higher node energy consumption, this might degrade data dependability [53].

5.10 Priority Based Slot Allocation to Sensitive Data in WBAN

Patients' data is categorized as sensitive, on-demand and routine data. Sensitive data consists low and high threshold values for vital signs such as heartbeat-based body sensors identifying lower threshold value and respiratory body sensor identifying a high threshold values based on high respiratory rate [54]. This is a body manager that collects data from any BMS and delivers it to a physician on demand. Temperature, blood pressure, and glucose levels are the daily data. Consequently, sensitive data require greater attention to access to the channel first compared to regular and routine information. Researchers will upgrade energy-efficient MAC protocols with intensive slot allocation schemes so that sensitive data can be transmitted with or without conflict.

6 In WBANS: Multiple Access Scheduling Schemes

In the MAC Protocols the BMS slot allocation activities are performed with the support of various scheduling schemes for multiple accesses (MA). MA schemes are available such as Slotted Aloha, TDMA, FDMA, CSMA/CA etc.

Scheduling schemes are used to maximize throughput, channel utilization, fast and reliable data transmission, minimize collision, delay, re-transmission and energy utilization. Existing MA schemes at the MAC layer are therefore classified into three main categories: scheduling, contention and hybrid based [55]–[36]. Such three scheduling access techniques helps to allocate channel to a patient's data's heterogeneous existence. In different research papers, the essence of patient data is categorized into three categories. The grouping of patient data into three categories is not justifiable, however, because the higher and lower threshold values and sensitive delays are not addressed in these classifications. Currently, the existing literature classifies the data of the patient into four to five categories [56]–[57]. The data of patient is transmitted to body coordinator using different access techniques as defined in the sub-sections below.

6.1 Schedule-based Slots Allocation

The TDMA and FDMA are schedule-based slot allocation mechanisms and are used for MAC Super frame structure slots of the Contention Free Period (CFP) [58]–[59]. TDMA scheme is a channel access method for shared-medium networks, it splits the signal into separate time slots to allow many nodes to use the same frequency channel, in which

each body node waits and transmits emergency information [60]. For example, node-1 and node-2 are routine data when transmitting the allotted predefined time axis. Node-3 simultaneously detects sensitive data during node 1 and node 2 data transmission. Node-3 will wait in this life-threatening scenario and transfer sensitive information within the predefined allocation time frame. Therefore, due to the long period of wait the performance of the schedule based scheme degrades as well as ruins the life of the patient, the predefined-based time frame allocation is not suitable for emerging genetic data [61] (Fig. 10).

6.2 Contention-based Slots Allocation

The contention based scheme like CSMA/CA scheduling access scheme is the most widely adopted because of its simplicity and infrastructure-free data transmission [58]. The slot distribution system for body sensors used by CSMA/CA is based on the first come, first served (FCFS) concept [62] and IEEE 802.15.4 MAC Superframe structure is configured for the CAP time frame. In duration of the CAP time, the BMS denies access to the slot (channel) and all BMSs are equally likely to access CAP slots [63]. The body manager

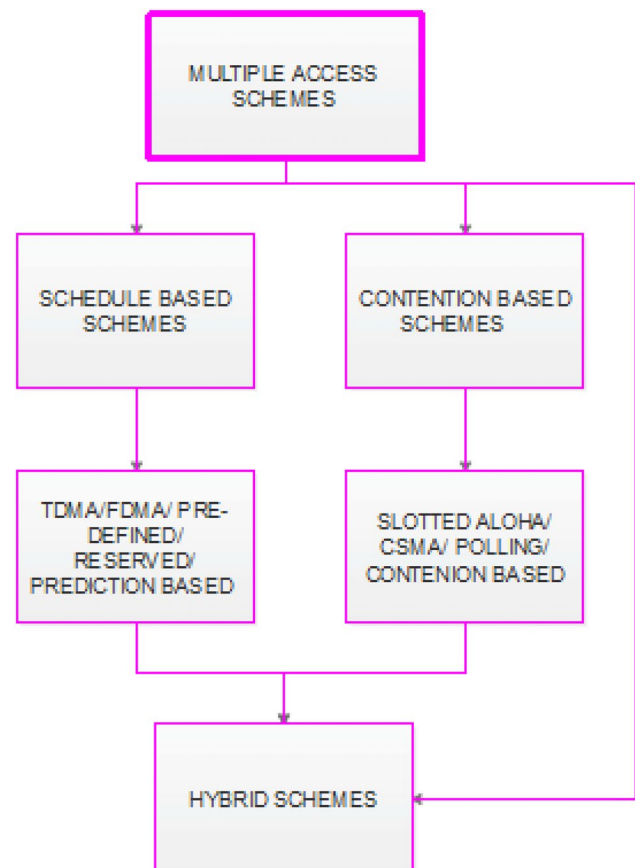


Fig. 10 WBAN: Multiple access Schemes

also allocates the CFP slots to those body sensors that gain access to the channel during the CAP time because of this difficult issue, collisions, retransmission of missing packets, increased delays with worse data quality and better energy usage by BMSs, for sensitive data this approach is not suitable [64].

6.3 Hybrid-based Slots Allocation

These slot allocations are the combinations of scheduling and contention based methods like TDMA combined with CSMA/CA [65]. Most of the researchers, used schedule based approach like TDMA-based approach for regular or routine data-based BMSs and for sensitive or critical data contention based approaches like CSMA/CA, Slotted aloha etc. whereas these types of BMSs send warning signals to the coordinator in the event of low or excessive vital indicators [55].

6.4 Comparative of MA Scheduling Schemes

Wireless data communication between body nodes and the body coordinator is designed with the help of scheduling access schemes [66]. Many MAC frameworks have been noted to implement TDMA with CSMA/CA access schemes to assign channel in CFP and CAP times. Higher energy utilization, low throughput, higher delay and data collision are the most difficult issues that degrade MAC protocol quality for BMSs to access channel during contention. Similarly, in their predefined time slots, TDMA-based BMSs are waiting and transmitting data. Therefore, Because of the conflict and longer waiting time, both scheduling access schemes are not appropriate for emergency information based BMSs. It [15] presents and compares the different types of network topologies, energy utilization, data rate, bandwidth, packet delivery and synchronization features of both scheduling schemes. Nevertheless, the usage of energy of the contention based scheme is higher than that of non contention based schemes.

7 Classification of MAC Protocol

This section examined MAC protocols for WBANs based on scheduling, argument, and MA scheme. The effect on key metrics such as data quality, collision, delay, latency, packet transmission ratio and energy consumption is investigated in MAC protocols studied under taxonomy.

7.1 Schedule Based MAC Protocols

Predefined or prediction-based slot allocation is used in schedule-based MAC protocols. Every BMS waits in the

predefined time slot or prediction-based for the transmission of data. A BMS loses data during the waiting period and consumes a higher energy that breaks down MAC protocol performance [81]. If both sensors detect sensitive data or threshold values, the preset or prediction based slot allocation procedure is not appropriate for sensitive data due to the poorer data dependability and increased latency. Many research attempts have been undertaken to tackle the difficulties highlighted. In the following, we shall discuss several relevant MAC protocols.

The authors [67] suggested HBD-MAC (H-MAC) protocol to improve energy efficiency of sensors node with TDMA approach. The heartbeat of human body is used for synchronization instead of periodic synchronization beacons, to get periodic timing information from a body coordinator by turning on their radio. The biosensor broadcasts a threshold value if the flow of blood fluctuates. The main limitation, it does not investigate the slot allocation if two or more vital signs detect threshold values simultaneously that is not suitable for conditions that endanger life.

The medical-MAC called Med-MAC [68] is a TDMA based protocol with feature of dynamically adjusted time slot algorithm. The sensor nodes for respiration, pulse and temperature are called class 0 due to low grade data, ECG, blood pressure, SpO₂, EEG are called class 1 due to average grade data and videos, medical images, EMG, capsule endoscope are called class 2 due to high grade data. For each time slot, the synchronization mechanism adds a Guard Band to allow the node sleeping mode between one or more beacon cycles. An Adaptive Guard Band Algorithm (AGBA) measures the GB to avoid collision. So that each body node has a specific time slot. Whenever a node detects unpredictable data then the beacon period is used. However, contention, synchronization and lack of critical data knowledge reduces MedMAC performance in terms of delay with low data reliability.

The proposed protocol [69] performs more better in terms of energy saving as in performance evaluation, comparison to X-MAC, B-MAC, WiseMAC and ZigBee (802.15.4) and demonstrates power efficiency in single hop scenarios. In TaMAC, the normal, on-demand and sensitive traffic is classified and for regular traffic, preamble scheme is utilized. while for on-demand and emergency traffic, wake-up radio mechanism is implemented. TaMAC does not consider bursty traffic load on more than one body nodes simultaneously due to sensitive data and considers periodic traffic. The time required for wake up response signal and beacon transmission results in higher delay when transmitting emergency traffic. In contrast, TaMAC only examines traffic that occurs on a recurring basis and does not concurrently acknowledge the burst of traffic due to emergency traffic, many sensor nodes may be affected.

The protocol suggested in [70] reduces energy consumption through the use of state switch with the main radio port

and wake-up radio port. By default the BSNs are set in sleep mode and wake-up by BC when they have the data to transmit or receive. The behavior of BSN and BC are different on the classification of data. PE-MAC classified the traffic into routine or normal and random or sensitive data. The limitation of [70] is that in the predefined time slot, either routine data or sensitive-based, waits and then transmits data. Due to the waiting interval, the MAC protocol's performance suffers due to a larger latency, reduced data dependability and higher energy consumption. The second limitation is that there is no distinction between critical values of vital signs in the suggested protocol.

The proposed protocol [71] is suggesting dynamic TDMA scheme to enhance packet delivery rate, channel utilization and to reduce duty cycle and power consumption which are basic requirements in WBANs. Basic Time Slot field (BTS) is dedicated for transmitting control packets and Dedicated Time Slot field (DTS) is dedicated for data receiving and transmitting. Both DTS and BTS are TDMA-based and collision-free, except for an optional Contention Access Period (CAP) field. It categorizes the data into regular, on-demand and emergency traffic. To maintain data reliability, reallocation procedure for sending failed packets to minimize the delay. The emergency and on-demand data transmission leads to delay due to overhead packets and re-synchronizations which are not suitable for life threatening situations.

The proposed system [72], a cloud-assisted MAC protocol, CLNC-MAC, has been designed to use network coding and ensure the eligible transmission of whole prepared data to the destination. In a cooperative network environment, it uses Random Linear Network Coding (RLNC) to guarantee that all BSNs' data is transferred to the destination without any issues through the network. This is done by cloud controller that acts as inter-mediator. On the cloud all the network coordination tasks are performed by relay coordinators and at the BSNs data operations are performed. Due to relays higher delay in data transmission may be possible which leads to threatening of patient's life.

The proposed scheme [22] facilitates moment-critical applications by enabling access based on TDMA in networks allowed by beacon. In this network, BC transmits data with the body nodes through GTS. To overcome the limitations of traditional standards, adaptive GTS scheme is proposed that will optimally use the resources. The end devices can request all types of GTS with duration T-short, T-medium and T-long respectively. The lifetime of network suffers and more energy is consumed in GTS request frame and beacon frame communication. Further, due to mentioned scheduling algorithm it lacks adaptability and Due to FCFS it might be possible that the sensitive data based sensors drop packets.

In this scheme [73] the Body Coordinator acts as master and each body node required a synchronization for data transmission. The traffic is classified into high and low

priority packets. Allocation of slots is based on FCFS so it might be possible the emergency data drops packets and it needs re-transmission. So it consumes energy and loss of packets in waiting of time slot. Moreover, the threshold value of vital signs are not taken into account. Due to the greater latency, these settings reduce the efficiency.

In paper [74], Two Round Reservation MAC, named TRR-MAC is presented IEEE 802.15.6. Priority of users and health severity index are used to categorize loads. To minimize energy consumption, for burst traffic two round reservation technique is introduced and one-round reservation for periodic traffic. Presents a unique bit sequence scheduling method for m-periodic data traffic, and create a token by employing a severity index to reduce the transmission delay of emergency data. Though, if the sensor nodes are not using the dedicated reservation slots as well, the other nodes will not be prepared to use those time slots.

In proposed protocol [75] a optimized framework is designed for high efficient smartBAN that minimizes the average delay of data transmission. An adaptive interbeacon interval (IBI) algorithm is suggested to specify the optimal IBI during the communication between the body sensors and hub. There is no adaptive channel allocation to emergency and on demand data as they have not given any priority. The results shows better performance in terms of energy savings and channel efficiency for periodic traffic. This scheme does not consider and investigate busy, on demand and emergency traffic which is the major requirement of WBANs.

The proposed protocol [37] is TDMA to ensure multi-beacon synchronization and contrast its performance with the CSMA/CD approach. The body sensor node has capacity to manage data and the communicate with body coordinator. A TDMA-based self organization scenario is designed to transmit data from wearable body sensors to master device based on synchronization signal. Periodic messages are used to divide the nodes into clusters. To avoid collision each cluster is identified with unique number managed by master device. The protocol consumes energy during synchronization, cluster creation and cluster release operations. There is delay in emergency data transmission and packet loss might occur.

A greedy strategy is proposed in [76], based on time slot allocation to reduce time complexity to solve the problem. Sensor priority is used to construct a utility function, available energy and sampling rate of the node. In this framework, highest priority nodes are arranged with better channel conditions to relay data frames under time slots. The objective is to optimize the data transmission process of all body nodes during a certain time by varying the time and length of transmission of each node. The primary goal is to optimize data transmission for each node in the network. But data gathered by sensors are not categorized and no low and high threshold values calculated.

In [77], to minimize inter-WBAN intervention, a priority aware price power control scheme based on the Stackelberg and Bayesian game models was proposed. Extensive simulation findings demonstrated that sensors based on the proposed PPPC strategy could change their transmitting forces based on priority levels, improving the timeliness and efficiency of sensitive information transmission and extending network life.

A novel nest-based scheduling algorithm (NBWS) is implemented in paper [78] to reduce the interference within inter-WBAN or intra-WBANs. The proposed algorithm is to group similar sensor nodes in a one group to make clusters. The graph colouring theory was utilised to schedule all groups to work to use a Time Division for Multimodal Sensor (TDMS) group scheduling model. The proposed NBWS algorithm performs better than the counterpart techniques in terms of collision, frequency propagation latency, system throughput, and energy consumption, according to the results.

In [79] paper, energy efficient HB-MAC protocol proposed PHY layer, C-BCC on ultra-wide-band, suited to low-power activity on the human body. HB-MAC, like many low-power protocols, divides time into random-access and GTS. It has been developed to operate efficiently under a universal clock dependent on heartbeats and allows lightweight devices to be more effectively integrated with inconsistent traffic.

A cluster media access control is designed to enhance the superframe length as per to higher priority traffic, whilst remainder of the data load is managed using node buffering (nodes-based buffering) [80]. The traffic is clustered on the basis of priority and the time in the superframe is dynamically allocated depends on the amount of buffered packets and priority.

As per literature, the goal of these protocols is an efficient multi-channel MAC that carefully plans message forwarding and leverages multiple channels to maximize parallel forwarding between adjacent nodes to avoid conflicts at the MAC layer. In fixed or scheduled slots, the packet delay and loss may occur due to buffer over flows which may degrade the WBAN performance. Further, creating network-wide schedules can be a daunting task and these schedules may need to be modified each time the network topology or network traffic characteristics change.

7.2 Contention Based MAC Protocols

TAD-MAC protocol [29] using CSMA/CA scheme, the authors present an adaptive algorithm in which every node has its own traffic status register (TSR). To transmit data each node must wait for beacon signal send by body coordinators. On successful and failure data delivery the recipient node responds '1' and '0' respectively. It save energy in

terms of overhearing, idle listening, collision and unnecessary beacon transmission but not suitable for WBANs due to preamble-based strategy. The nodes having emergency data has no priority and no traffic classification has been considered.

The Priority-based Load Adaptive MAC (PLA-MAC) protocol [57] to minimize energy consumption with traffic load estimation based prioritization. That classifies the data of the patient from higher priority to low priority as Critical Packet (CP), Reliability Packet (RP), Delay Packet (DP) and Ordinary Packet (OP). All nodes are required to contend in the CAP period at the start of channel allocation. Emergency sots are given to emergency based sensors, but they need clear channel assessments (CCA) to avoid collision during transmission. However, protocol consumes higher sensor energy during the contention period and does not distinguish threshold values of sensor nodes. It reduces MAC's performance With a higher latency and with low reliability of data that is not suitable for sensitive data.

To classify the data R-MAC protocol is proposed based on Radio Frequency Identification (RFID) [82] using CSMA/CA access. To transmit sensitive data, the node will contend and wait for clear channel access. CFP slots are allocated by the coordinator node to those nodes that receive channel access during the CCAP period. The suggested protocol reduces MAC performance with regard to higher delay and collision as each body node claims to access channel during alarming situations.

Fuzzy Control Medium Access (FCMA) [83] implements phases of acquisition, fuzzy logic control and implementation to handle regular and sensitive data by CSMA/CA technique. To gather data form sensors data acquisition phase is deployed. When the patient's data is received by a body coordinator, It utilizes fuzzy rules to determine whether CAP or CFP periods should be allocated. During the CAP time, the fuzzy rules prescribe the CW Size for priority information and data frequency. The method suggested is appropriate for sensitive data, but during CAP sensors consume a higher amount of energy during the decision-making process, sensory data is delayed in allocating slots to sensors.

In paper [84], a priority slotted Aloha scheme is introduced for non-saturated networks with finite body sensors together with Discrete Time Markov Chain. It tries to resolve the conflict by minimizing the chances of re-transmission in the following alternate slots, less bounded to a minimum. In this paper, in the existence of a finite number of nodes, present an analytical model to estimate the unsaturated throughput of this protocol. The model is validated for simulation purposes. With higher data collision, the data quality degrades if more BMSs claim to access a slot at one moment. This scheme does not, however, rely on the information of the client, and the energy consumption of BMSs

during contention is high. The disagreement over the allocation of slots is the same problem as found in this report.

To improve QoS for the emergency data, this paper [85] proposes multiple level-based design. Priorities given to various emergency data collected by emergency nodes. Each node send request for the time slot to router and then router update its priority by allocating time slot to node. Moreover, If the router priority changes, a new GTS request is sent to the body coordinator. More energy is consumed in process of updating the time slots on priority basis. Further data delay and collision may be occur which is not suitable for WBANs.

The authors [38] proposes a Low-delay Traffic-Aware Medium Access Control (LTA-MAC) protocol to enhance performance of the MAC protocol in terms of data collisions, delay and data reliability. In IEEE 802.15.4 standard, during CAP period the BMS repeats the values of each round of contention in the next rounds of contention which leads to collision, data loss, delay and higher energy consumption. To overcome this problem, Contention Differentiated Adaptive Slot Allocation CSMA-CA (CDASA-CSMA/CA) for scheduling access scheme is proposed, where guard bands are deployed between channels to eliminate overlapping of the designated channels. Simulation results demonstrate advantages of the proposed schemes over the IEEE 802.15.4 MAC and CSMA/CA scheme in terms of success rate, packet delivery delay, and energy consumption.

A [86] priority aware adaptive MAC-frame payload tuning mechanism is implemented and proposes adaptively tuned MAC protocol (AT-MAC) for WBANs to maximize the probability of successful packet delivery or reliability of the associated sensor nodes based on real-time situation. To maximize the reliability of the sensitive data, a markov chain-based analytical approach is used for different user priorities of traffic. The results are evaluated on saturated networks and non-emergency data is ignored which degrades the overall performance of Mac framework.

In [87] authors propose a multi-channel MAC protocol towards energy efficiency and low interference (EIMAC).

Channel mapping mechanism to categorize the states of channel. To differentiate low and high priorities channel selection strategy is introduced and considering data volume, residual energy and user priority. The size of contention window is small for high priority nodes. Authors try to reduce the contention period and congestion in single channel and decreases the delay up to some extent with increase in number of nodes as compared to IEEE 802.15.6. However, higher energy consumed when packet loss and delay occurs for low priority nodes.

The protocol TPCAT-MAC (Traffic Priority-based Channel Access Technique) is proposed in [88] article

utilising IEEE 802.15.6 to reduce the transmission delay of essential data packets and to alleviate issues with other priority nodes during most of the back-off phases. A Low Threshold Criticality-based Adaptive Time Slot Allocation Algorithm (LT-CATA) and High Threshold Criticality-based Adaptive Time Slot Allocation Algorithm (HT-CATA) were introduced to reduce the priority slot conflicts between the different types of user priority nodes and low and high threshold traffic from the same. The implemented protocol provides better overall performance of delay, energy consumption and utilization by reducing channel access time for both critical and non-critical nodes.

In [89] eTA-MAC protocol is proposed to provide priority to emergency traffic. The proposed model is offering three steps for the accomplishment, on basis of categorization of data packets eTAMAC protocols TCPCSMAC/CA (Traffic-Class Prioritization-based slotted-Carrier Sense Multiple Access/Collision Avoidance (TCP-CSMA/CA)), ETCPSCMA/CA (Emergency Traffic Class Provisioning-based slotted-CSMA/CA) and ETAC-SMA/CA schemes can work together to improve throughput and reduce latency and energy consumption. The ETCPSCMA/CA method delivers sporadic emergency traffic instantaneously generated by one or more BMSNs with minimal delay and packet loss. Perform this task while recognizing normal CAP traffic. Then the ETACSMAC/CA approach uses normal traffic BMSN energy storage to create a balance of energy and throughput for sporadic emergencies. In [90] article, to minimize delays, collisions, and energy usage, the authors proposed a two-step protocol called the PCCMAC protocol. In the first step: To reduce collisions, use run-time statistics to prioritize sensors and allow access to channels only when potential nodes send data. increase. In the second stage: The size of the CW (Conflict Window) is estimated using the hit rate and queue length, which accurately mimics the current channel state.

CAP latency is high and can affect WBAN performance. The authors of [91] proposed TPLBS (Traffic Prioritized Load Balanced Scheduling) Protocol. It focuses on the waiting period of data packets waiting in a queue before sending them to the access point. According to importance, data packets are sorted and split into 4 queues. For on-time delivery of emergency traffic, the TPLBS algorithm is being proposed to ensure that critical data packets are properly buffered.

According to the mentioned literature it is concluded that, the main key issue is collision. Although the problem cannot be completely resolved, these types of protocols can reduce collisions by random backoffs. These protocols are scalable without the constraints of precise time synchronization. However, it increases the protocol overhead.

7.3 Hybrid Approaches Based MAC Protocol

In hybridized approaches, combination of contention and schedule based schemes are implemented. In most of the research work, schedule based scheduling like TDMA slots are used for sensitive-based BMSs, and Contention based Like CSMA are used for ordinary and regular BMSs.

The authors suggested Medical Emergency Body MAC, named as MEB-MAC in [92] to enhance the performance of WBANs. The Exclusive Access Phase (EAP) therefore is eliminated and short frames are used for emergency transmission, that also increases the overall results. It is claimed that it is not possible to use both RAP1 and RAP2 for emergency transmission as there is a risk of failure or collision. The supervisor implements several listening windows to prepare for emergency transmission during the controlled access period without informing the body coordinator using TDMA. Emergency sensors can send their information in listening windows. It can also guarantee reliability and optimize access time. The length of the phases of access is flexible and the application criteria are specified. Traffic is graded according to intensity and a timer for back-off is set accordingly. The body coordinator takes care of all the emergency data demands. Furthermore, the protocol consumed higher energy which reduces the network life-time. This is primarily due to the elimination of EAP and MAC's complex access time.

The body coordinator initiates the synchronization by transmitting the beacons as the start of communications in P-MAC protocol [93]. The superframe starts with a preamble for synchronization followed by Down-Link (DL) and Up-Link (UL) sub-fields. The subfield Down Link determines the time of the occurrence of the physical layer in the subframe Connection down, which can be unicast or transmitted. The UP Link includes each node's information and determines each node's probability of transmission. TDMA mode is categorized into dual parts; Time Slot Reserved for ET (TSRE) and Time Slot Reserved for Regular Traffic (TSRN) to meet emergency as well as normal traffic requirements. The normal traffic can use uplink traffic control and emergency can use both emergency and normal traffic link. The CSMA mode is divided into CAP and subdivided into two control streams, Uplink Emergency Traffic Control (UCE) and Uplink Normal Traffic Control (UCN). In order to improve energy efficiency, the P-MAC protocol enables sleep mode. Upon good data receipt, the body coordinator sends ACK to the body node, and it can go to sleep again. The idea presented in this protocol is theoretical and the performance is not measured by any concrete results. The superframe's overall structure is quite complex, The acceptance of traffic changes could therefore be small. The practical implementation of this protocol is also a challenging task in itself.

The authors [94] proposed modified frame structure (MFS-MAC), which is proportional to a number of nodes in WBAN, to reduce energy consumption. In this review, the body coordinator establish communication link among body nodes. The design of an MFS-MAC superframe merges phases of Type I and Type II and transfers it to the end of the superframe to eliminate polling. Secondly, EAP2 and RAP2 obey EAP1 and RAP1. If transmission happens at EAP1, the node will sleep in order to conserve energy after transmission. The downside of this method is that only when there is a node with data to be sent, the coordinator wakes up. The research emphasis on body coordinator so, the resulting performance is comparatively low and the access delay compared to other protocols is also high.

To achieve data reliability authors [62] propose Ada-MAC protocol for real time communication with CSMA/CA techniques. To reduce energy cost after completion of data transmission each BMS switches off its radio during CFP. For body node that has burst or emergency data to transmit uses time-triggered mechanism as protocol allocate appropriate number of time slots and by using CSMA/CA other nodes in the left slots will transmit their data. In case of sensitive traffic to avoid collision they utilize prioritized back-off. It shows good results in terms of delay and packet loss ratio. For different type of traffic prioritization it uses multiple queues like IEEE 802.11e that is not feasible approach for WBANs.

The Human Energy Harvesting Medium Access Control (HEH-BMAC) protocol [106] is specified for battery charging purposes of body nodes from a person's harvesting energy, use sensitive data, ID-polling time slot and periodic data. A supervisor assigns monitoring intervals to each node in order to calculate the Inter-Arrival Time (IATBN) and energy harvest speed (KEH). The body coordinator transmits the Control Packet (CP) to nodes while the nodes determine their data packet transmission threshold values (X_i). If the node does not receive the packet from the CP, the body coordinator will wait for the predefined interval and send it back with the threshold values being changed. The data packet collision occurs, when more than one node has the critical data at same instant and ready for transmission. The this scheme[95] an Interrupt Mechanism (I-MAC) hybrid medium access control protocol is proposed to improve the efficiency of energy and time slot allocation and at the same time satisfy the data delivery latency requirement. Unlike existing hybrid MAC protocols, to avoid unnecessary beacons, a superframe structure with a longer length is adopted. Most of the time slots are distributed to nodes with periodic sources of data. Fast interruption slots are inserted into the superframe to relay the urgent information and to ensure that these data requirements are met in real time. The coordinator will split the running superframe and start a new superframe during these interruption slots. Only when there are more

data to be delivered is a contention access period (CAP) activated. Results show the adequacy of the proposed MAC protocol with low urgent traffic delay in WBANs. Waiting time and delay of data transmission increases for all the other nodes when the body node has longer length data to transmit in one time slot.

The authors suggest [96] a hybrid priority-based MAC protocol for WBANs powered by a disagreement over the reservation mechanism to reduce the delay in the delivery of emergency information. For sensitive data, time slots are dedicated and Other nodes can also also transmit data over dedicated time slots by using traffic adaptive approach. In regular data, by increasing the number of sensor nodes, the average delay is increased.

In this paper [97], to minimize energy utilization a hybrid prioritization scheme is proposed. At physical layer and MAC layer body nodes are prioritized, depending on the transmitting power. Nodes associated with different priorities at the MAC layer by assigning different window sizes to them. Furthermore, body nodes having similar priorities are further prioritized based on transmitting energy. At the outset, the loss of the route is determined by transmitting the beacon to the physical layer. Node determines when to use path loss to engage in the access to contention based on predefined threshold values. The nodes with high energy transmission engage in the connection to contention.

It therefore limits the number of participants, leading to a lower number of collisions. In addition, in the case of an odd number of retransmission errors, this scheme decreases the back-off time. For each priority class, this scheme requires prior knowledge of node transmission powers. Advance node awareness in the context of high density and diverse nodes is a difficult task.

The authors presented [98] the Channel-aware Polling-based MAC protocol (CPMAC) to overcome the existing shortcomings in sleep-wake-up scheduling mechanisms used in traditional static sensor networks that could be very inefficient in WBANs. Make it possible to maximize our CPMAC's energy efficiency by adjusting the superframe polling cycles to complex traffic and channel fluctuations. This model only wakes up sensors and this simple strategy could trigger body sensors to transmit when the channel is powerful enough for sensor networks, but it could be relatively inefficient to ensure fast and reliable data transmissions.

In [99] the authors propose hybrid DTDMA/DS-CDMA protocol that incorporating the benefits of dynamic TDMA and DS-CDMA, in order to adapt BMSs variations in traffic and different requirements for Dynamic TDMA scheme is used for regular traffic to avoid collisions, idle listening and overhearing. On the other hand, DS-CDMA is used for sensitive traffic where nodes can concurrently transmit their data across the network. This results in low induced

delays even in environments of high traffic load. A new urgent-conscious hybrid DTDMA/DS-CDMA protocol that addresses emergency and periodic traffic conditions to address body node traffic variations. This benefits from DSCDMA's high delay efficiency in high traffic, and from DTDMA's high energy efficiency in regular traffic. The proposed scheme will be tested in terms of delay, packet fall rate and energy consumption. Development reveals that it outperforms the other in terms of packet drop percentage and delay for various number of nodes carrying emergency data as well as different payload sizes. It also provides the highest energy efficiency during periodic monitoring, thus modifying energy efficiency in emergency situations by allocating spreading codes only to nodes that carry emergency information.

The authors proposed hybridized [100] the method that is implemented on slotted aloha, the CSMA/CA and the TDMA standards for transmission to differently considered energy balance in body nodes and body coordinator and instantly transmission of critical data to the body coordinator regardless of any time constraint. The proposed scheme loads the inevitable energy consumption in the coordinator instead of the nodes to extend their life. The major advantage is emergency data can be transmitted simultaneously with high priority with maintaining a periodic data transmission. The proposed mechanism could achieve the enhancement of existing mechanisms despite the fact that a few probably won't be improved. The latency of periodic data transmission increases and the energy efficiency of body coordinator decreases due to enough load which reduces overall throughput and degrades the performance of WBANs.

The suggested protocol [101] uses a hybrid scheme by using CSMA/CA with collision avoidance and TDMA schemes. The load transfers to main station or coordinator to improve energy efficiency of body nodes. The major task of body sensor is to send data to body coordinator. Allocate the bulk of slots to body nodes for transmitting data through a novel hybrid scheme CSMA/CA-TDMA. If body coordinator has to transmit information to the body node, it sends a switch beacon after the guard time interval (GTI) at the end of the receiving cycle of a PS. The GTI is used to avoid duplication of time slots. The protocol uses a novel TDMA scheme to save power for BSNs during the receiving time of body coordinator. This needs to improve the protocol's scalability because there will be more and more body nodes in the process.

In this paper [102], to improve the WBANs performance, it combines both IEEE 802.15.4 and IEEE 802.15.6 standard protocol. IEEE 802.15.4 is more efficient for sensors with minimum data rate requirement, for sensors enquiring high data rates, the use of IEEE 802.15.6 is more advantageous. It provides sensor communication mechanism between hybrid

and protocol switching algorithm according to the data frequency and the time frame metrics.

For adaptable network due to fluctuations of business traffic, the authors of [103] offers an adaptive MAC (A-MAC) technique for WBAN based on IEEE802.15.6. The contention access phase is a three step process, with the length of the sub phase being dynamically adjusted according to the data priority. The random data that competes successfully transmits data directly, and the periodic data that compete successfully transmit data in the allocated time slots of the non-contention access phase. Finally, all nodes compete for access to the same channel when they are not in contention. During the non-contention access phase, just one node sends data in each time slot, preventing collisions and ensuring high dependability and low latency. Time synchronisation between the coordinator and sensor nodes is necessary, which will result in some extra energy usage. There is no need to utilise additional energy during the non competition access phase since long-distance communication happens.

For implanted biomedical sensors the authors of [104] proposed a Thermal aware duty-cycle MAC called ThMAC—protocol. It introduces a superframe structure that uses both contention-based and contention-free channel access methods to accomplish QoS provisioning by utilizing both CSMA and TDMA techniques. Through an emergency data management mechanism, ThMAC enables the dependable and timely distribution of occasional emergency data. In regard to Quality of Service (QoS) demands the traffic is well categorized into emergency, reliable, normal and delay constrained. ThMAC attempts to decrease node energy depletion in order to extend network lifespan. Finally, ThMAC handles emergency traffic management, attempting to provide Emergency traffic with the least amount of delay while maintaining high dependability.

In the critical review paper [39], the realistic simulation is performed on NS-3 simulator for health monitoring to evaluate average delay, throughput and power consumption. Authors incorporate key structures such as an association, connection-oriented link distribution, channel hopping, and a two-hop star topology extension, in contrast to current simulation schemes for WBANs.

In [105], hybrid MAC protocol proposed where two main phases, namely the contention phase (CP) and the transmitting phase (TP). The slotted ALOHA protocol is used as a contention method for the reservation of transmission slots during the contention process, while the TDMA protocol is used for transmission during the transmission phase. To show optimization Monte Carlo simulation was performed on MATLAB which is not practically suitable for WBAN scenarios.

As literature shows that the combination of scheduled and contention based protocols give better performance for heterogeneous data as for emergency or sensitive data scheduled

slots like TDMA are used and normal and non-emergency BMSNs performs contention to access the channel.

8 WBAN MAC Protocols Performance Metric

There are several simulators and emulators used in the WSN and its applications such as WBAN research. The most common ones are TOSSIM emulator [107] can handle thousands of simulations as a very simple and powerful special network emulator, NS2/NS3 are network simulators version-2 and version-3 respectively. NS2 is act as simulator only whereas NS3 can be act as the emulator that it can connect to the real world [108]. The JavaSim (previously known as JavaSim) [109] simulator, an emulation node that executes the TinyOS operating system. It is the JavaTM- basic simulator which was generated in the after NS2 mode to resolve the scalability problems facing NS2. In terms of functionality and model precision, Castalia simulator is much superior to these and other WBAN Simulators [110]. Designers may carry out a full set of simulations with only Castalia's attention, varying from various scenarios and prototypes to different scaling parameters. Some essential and foundational topics of Castalia are discussed in the following section.

8.1 Summary of Castalia Model

This section provides us with a concise description of the models that contribute to the simulation findings discussed in this paper. See the Castalia User Manual for a more detailed description [110]. Castalia was built on OMNeT++, a framework for the computer network simulation which provides essential machinery and tools for the writing of discrete-event simulators. Castalia is a system of wireless networks of sensors (WSNs), the body areas and, in particular, the systems of low-power embedded devices. It can also be used, since it is extremely parametric, to test different platform features for particular systems and can also model a variety of platforms. OMNeT++, a fantastic platform for setting up event-driven simulators, enables Castalia partly to be scalable, stable and efficient. Such standardized and consistent structure for validating a specification first-order before it is introduced. By predicting temporal shifts and cumulative route losses dependent on the human body variants, Castalia is by far the most accurate BAN network simulator. Various sensor nodes or nodes are not directly connected to each other as shown in Fig. 11. In order to send packets, nodes use the Wireless Channel (WC).

The arrow indicates the communication from node to device. The Wireless Channel (WC) determines who can accept the packet. As shown in Fig. 11 nodes are monitoring

Physical Processes (PP) and connecting to each other with PP. So there are number of Physical Processes (PP) known

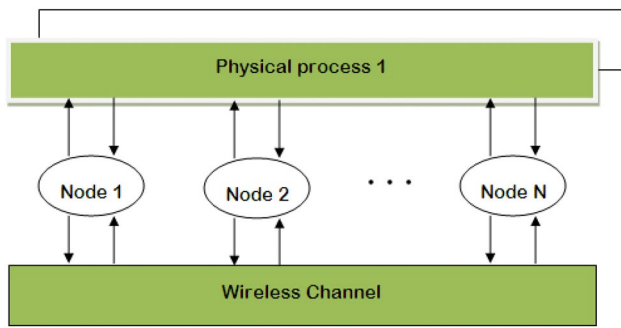


Fig. 11 The Higher level Module Connection in Castalia [111]

as sensor device. Every Physical process should have one module to hold the value like reading of temperature, pulse rate etc. This is done by having nodes sample PP in space and time in order to get their sensor readings. One node has multiple processes representing the multiple sensing devices. The common method in the production of sensed data is to feed random numbers into nodes, or at best to feed sensed data traces to each node. Castalia adopted a general physical process model to supply data to the node sensing system. The **inputType** parameter specifies the physical process will be described by the model given above or by direct assignment. The format is **double inputType = default (1)**. The generic module defines the physical phase if the value is set to 1, whereas 0 for direct assignment. Castalis defined two types of PP:

- Customizable Physical Process: Define the behaviour of the source depending on the location by setting the values of parameters like **string source 0 = default("0 10 10 30.5; 5 10 10 45; 12 10 10 7.3")**; The string defines the form: "time pos x pos y value; time pos x pos y value; time pos x pos y value;...".
- Cars Physical Process: This simulates the behavior of cars driving through a certain stretch of lane (for example a bridge). In the simulation space, the path is described by two points with x and y coordinates. These points are determined by the module's four parameters: (**point1 x coord, point2 x coord, point1 y coord, point2 y coord**). Where a car is required, it is arbitrarily allocated to drive from point 1 to point 2 or in the opposite direction. When car reach the destination, the car vanishes from the simulation area.

The whole framework of Castalia is expressed in the directory hierarchy within the source code. Every module matches to a directory where .ned file that describes the basic model, input/output gates and parameters of module is always present. There is configuration file called .ini file that applies or re-sends the variables to a value other than

their default value. Modules and messages are the basic concepts of Castalia. The basic execution unit is a simple module. It reads or executes a set of software according to the communication from other devices. The code will indicate that the interactions are changed and can send (or schedule) new messages. Castalia supports the creation of own systems and applications by identifying suitable abstract groups as depicted in Fig. 12. The node is composite module. If module is composite their are sub-directories to define the sub-modules. Each module as well as sub-module has .ned file that define the input, output and parameters. The user can modify the application, MAC, routing and radio modules easily.

Castalia's core characteristics are:

8.1.1 Wireless Channel Modeling

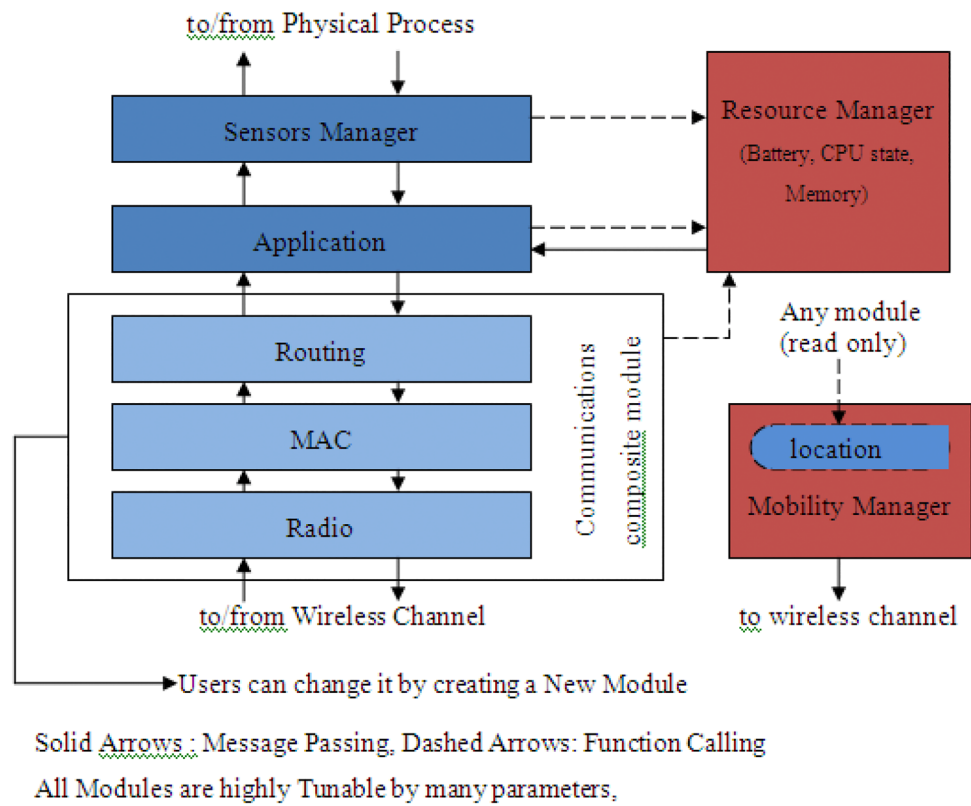
In general where mobile nodes, an evolving structure is taken into account, including of BAN, the wireless channel is a very difficult design. With respect to wireless interface, Castalia is the most versatile WSN and BAN simulator. The term straight is being used here in the sense that perhaps the simulator guarantees that the various essential functions of the wireless channel can be identified. Another critical catalyst has emerged from the latest development in BAN and the next IEEE BAN standard. Extensive experimental beds, for example, produce tens of millions of NITCA samples. These calculations are described to provide reliable predictions for the normal loss of body direction as well as for the temporal behaviour of the stream in particular.

8.1.2 Radio Modeling

In Castalia, three radio files are defined under **Simulations/Parameters/Radio/** which follows a specific format. These are: BANRadio.txt, CC1000.txt and CC2420.txt. BANRadio describes the Narrowband (NW) radio proposed in the IEEE 802.15 Task Group 6 documents. Both the CC2420 and the CC1000 define real radios of the same name by Texas Instruments. The parameter **RadioParametersFile** of the Radio module points to this file. Radio parameters are controllable by the processor. Each node should have its own radio. It means the number of nodes are equal to number of radios. Some below mentioned parameters are dynamically set:

- Radio State
- Reception mode
- Sleep Level
- TxPower output
- CCA threshold

Fig. 12 The node composite module



– Carrier Frequency

8.1.3 Routing Modeling

Castalia 3.2. provides mul- tipathRings and bypassRouting routing protocols. Rings with different routes are referred to as multi- pathRings. Routing nodes are not assigned a special parent. During initialization, a node only receives a level number or ring number. Stage 0 is assigned to the first packet sent from the sink. The procedure is replicated, with each node adding one stage of the recipient packet. Both linked nodes would fi- nally have a level number. When a node needs to transmit a packet to the sink, it broadcasts it with its level number attached rather than transmitting it to a single node. It will be rebroadcast by every node with a lower level number. The operation is repeated before the sink is hit. The sink node is decided by the application or defined in .ini file by using parameter **isSink** takes values true/false. As a consequence, if multipathRings are used, the application must also specify the **isSink** parameter. such as **SN.node[0].Application.isSink = true** The algorithm could be more stable than single route algorithms, but if traffic increases a certain, con- gestion will degrade results. On the otherhand the bypassRouting does not support any routing. Sim- pleTreeRouting was introduced in Castalia versions 1.3 to 2.3b. Although there is less support for routing in Castalia

But still many researchers implemented their own routing protocols in Castalia due to its realistic nature [112].

8.1.4 MAC Modeling

The Medium Access Control (MAC) is the most effective framework for packet distribution and energy efficiency. There are there- fore a separate module defining the MAC Protocol, an important part of node behaviour. For sev- eral factors, mentioned WBAN topologies are a challenge. First, because of the existence of the data rate vector, the BMSNs are confronted with the problems due to low ener- gies and processing power [50]. However, traffic loads sometimes increase due to emergencies. The BMSNs still provide high-rate traffic for de- tecting ECG signals in real- time. Second, low-load traffic applications typically need the lowest time energy consumption and high-load traffic applica- tions generally need a high packet distribution ratio with minimal delay [50]. Of this purpose, energy efficiency mainly be taken into consideration in the event of low traffic demand, while the energy-to- packet ratio of heavy traffic should be equalized. Third, internal and external interference and insuffi- cient channel capacity impede the execution of MAC operations through an erratic and immoderate load. For WBANs to achieve high energy consumption and the packet distribution ratio at least at the minimum delay, a dynamic and scaled MAC protocol is therefore essential. As

MAC layer is interconnected with physical layer and to show the dependency on the radio, the data rate, Phy overhead and carrier sense operations are defined in In Castalia-3.2 total five MAC protocols are implemented. In this paper, In the next section, explored four major MAC protocols.

8.2 MAC Modeling

- (1) **TunableMAC Protocol:** A duty-cycled and contention Based MAC, exposes a lot of parameters to a user and the modulation implementation. The behaviour and attitude of a simple CSMA/CA MAC protocol also comes from TunableMAC [113]. However, protocol was designed to ensure broadcast contact (i.e., no unicasting). Ac- counts, control packages for Request to Send (RTS) and Clear to Send (CTS) are therefore not permitted. In Castalia by default the parameters are act like non-persistence with no duty cycle that means if channel is busy then the node will wait for some random amount of time and again sense the data. Ideally, the button should be notified immediately when the channel is free. But, in fact, the radio does not provide this notification function, so the node must do poll the radio to find the channel [114]. In TunableMac, after every 0.128 ms the node is poll the radio. But in case of P-persistence, it works like 1-Persistence CSMA by keep polling the channel until found free. When it found free then it sends data with probability P. To save energy consumption, reduce listening time of node by enabling the duty- Cycle, its fraction of time in which the node sleeps [115]. Many parameters can be change by users in .ned file(**src/node/communication/mac/tunableMac /TunableMA C.ned**).

Its persistence and backing off strategies can be fine-tuned [116]. Another essential feature is to duty cycle the radio to send a sufficient train of beacons during each data transfer in order to wake up possible receivers. Some important parameters with description are listed Table 9. **Dutycycle** and **listenInterval** are one of the energy parameters since they are related. Cycle starts again after each listen and sleep interval. In Dutycyle method, **beaconIntervalFraction** parameter is used to wake up the nodes before data transmission. The parameter **probTx** perform the operation with a certain probability when a data has to be trans- mitted (or retransmitted); often protocols presume this to be 1. The number of times the data need to be transmitted is defined by **numTx**. Obviously, the higher this parameter is set, the more energy is used, but the higher the efficiency (number of nodes reached) can be obtained. To prevent collisions be- fore transmitting data, wait for a random period of time first, then sense the channel to determine availability

Table 9 TunableMAC.ned: Parameters

Parameter	Description
dutyCycle	For (1dutyCycle) a fraction of the time the button is in sleep mode
listenInterval	This is the length of time the node continues to listen for each cycle
beaconIntervalFraction	The amount of time to send a train of beacons before transmitting data
probTx	Data Transmission with Probability
numTx	How many times should data be transmitted?
randomTxOffset	To avoid collisions, first detect the channel during a random interval
reTxInterval	Time interval between retransmissions
backoffType	Specifies how to set backup interval as in CSMA, if the channel is not clear, MAC will be back offs
backoffBaseValue	how long to backing off with each form of backing off
CSMApersistence	Determine the existence of CSMA Default (0) means CSMA is non-persistent CSMA
sleepDuringBackoff	Node go to sleep instead of waiting

by setting the value of **randomTxOffset** parameter [117]. If no acknowledgement is received then after how long time the data should be re- transmitted is defined by **reTxInterval** parameter. The **backoffType** and **backoffBaseValue** are channel sensing parameters that are used to evaluate the internal back off when the channel is not free in the CSMA technique. To determine the persistence of CSMA, **CSMApersistence** parameter is used, this parameter decides whether to sense the channel continuously or after some random amount of time. If working with duty cycle technique then node sleeps during backoff interval by setting value of parameter **sleepDuringBackoff**. Many MAC parameters capture the data from Radio-module. The default values can be changed in .ned file of Tun- ableMAC. CSMA technique is not concentrate on energy efficiency [118].

- (2) **TMAC MAC known as Time-Out MAC** is an enhancement of S-MAC(Sensor-MAC) allowing extendable active times) [119]. For WSN, T-MAC is a common MAC, utilizing several techniques to maintain low energy consumption (utilizing vigorous duty cycling and synchronization) while attempting to maintain high performance (for example, packet delivery) with the duty cycle changed in line with the traffic requirements [120]. The main Characteristics are as follow:

- Transmit all messages in burst of variable length and sleep between burst.
- Node sleeps while other sending data.
- Synchronization is similar to Sensor-MAC protocol.
- RTS/CTS/Ack/SYNC schemes are implemented

User can characterize the sizes of TMAC (SYNC, ACK, RTS, CTS) control packets and maximum data size and overhead frame sizes. Many parameters can be changed by the user in .ned file located at **src/node/communication/mac/tMac/TMAC.ned**.

Many parameters can be tuned as shown in Table 10. The **maxTxRetries** is used to define the number of transmission attempts for unicast transmission. The length of each nodes' frame period (in milliseconds) is defined by **frameTime** and it is important to note that this relates to the length of the entire frame, not just the active or inactive phases. The length of the contention interval (i.e. the time between randomised transmissions), in millisecond is defined by **contentionPeriod**, for every transmission request. This parameter's main impact is to prevent transmitting interference

from nodes in the network. To set the listen timeout duration **listenTimeout** is used for nodes to sleep to conserve energy. For ACK packet node has to wait, to set that duration **waitTimeout** is defined. After that time it is assumed that packet is failed. The **resyncTime** option specifies how much time elapses between broadcasting synchronisation packets (in seconds). The **allowSinkSync** option allows 'sink' nodes to skip contention intervals while establishing a network synchronisation plan, allowing for faster synchronisation. To turn on and off the RTS and CTS packets, the boolean parameter is set called. **useRtsCts**. To make sure that the MAC remains active for atleast 15 msecs, the **conservativeTA** is set true otherwise false. There are numerous ways in TMAC to keep the packet delivery ratio high by using less energy. The Castalia module provides the functionality of the TMAC protocol's forefather, known as S-MAC.

- (3) **IEEE 802.15.4 MAC: ZigBeeMAC** the standard for Wireless networks with low capacity, but it is not popular in WBANs. Castalia simulator implemented the core features of MAC layer, apart from its physical layer. In Castalia More specifically implemented are:

Table 10 TMAC.ned: parameters

Parameter	Description
maxTxRetries	The number of transmission attempts for a single unicast packet, including RTS packets
frameTime	The duration of each frame period for all nodes (in milliseconds)
contentionPeriod	The duration of contention period for all nodes (in milliseconds)
listenTimeout	To avoid interference contention interval in msec
waitTimeout	Wait the duration of the timeout (in milliseconds) for a response from another node
resyncTime	The interval (in seconds) between broadcast sync packets
allowSinkSync	Allow TMAC to pull information from higher layers to check current node status
useRtsCts	Allows to close RTS and CTS packages
conservativeTA	Make sure the MAC remains active for at least 15 ms after any activity on the radio

Table 11 IEEE 802.15.4:Parameters

Parameter	Description
enableSlottedCSMA	Slotted version with CSMA algorithm enabled
enableCAP	Allow the nodes to transmit Data packets in the CAP period of the frame
isFFD	Is the node a fully functional device?
isPANCoordinator	Is the node a PAN coordinator?
frameOrder	How long does the active part of the frame last during listening and sending?
beaconOrder	How long the full frame lasts (active and inactive)
unitBackoffPeriod	Specify how long the unit was used for the backing off
macMinBE & macMaxBE	When calculating the backoff time, this is the minimum value and maximum value that the backoff index can take respectively
macMaxFrameRetries	The maximum number of retries before the packet is considered lost and also the upper layer is notified
minCAPLength	The minimal period of the CAP duration while the use of GTS, described through symbols
guard Time	Handles time synchronization in milliseconds

- Guaranteed Time Slot (GTS) functionality
- CSMA-CA functionality (slotted and unslotted)
- Enable PAN (automatic association) for associated beacons
- Direct data transfer mode

As shown in Table 11, many parameters defined in 802.15.4 MAC can be tuned by the users in .ned file. To enable the slotted version of CSMA **enableSlottedCSMA** parameter is set as True. To define the CAP period of a frame it allows nodes to transmit data packets, defined in **enableCAP**. The parameter **isFFD** checks whether the node is a fully functional device. To distinguish the coordinator node **isPANCoordinator** parameter is defined. The **frameOrder** parameter is the active part of frame when radios are either transmitting or listening during communication. To define the length of full frame **beaconOrder** value is set. During CSMA/CA the backoff strategy is used in the protocol by **unitBackoffPeriod** parameter. To calculate the backoff time **macMinBE** and **macMaxBE** parameters are set. The **macMaxFrameRetries** set maximum number of retries during transmission, the packets may be lost or failed to deliver. During GTS slots the **minCAPLength** is the minimum length of CAP period. For time synchronizing, the **guardTime** parameter is defined in msecs. This protocol is appropriate for packages that require a minimal data rate and a long battery life in order to feature properly [121].

- (4) **IEEE 802.15.6 MAC: Baseline BAN MAC** Due to its versatile properties, such as fast data speeds, short range, low power consumption, different frequency bands, and access modes, IEEE 802.15.6 is highly popular in contrast to its predecessor. To find the module description users may find **src/node/communication/mac/baselineBanMac/BaselineBANMac.ned**. In table 12, the first three variables are **macMaxPacketSize**, **macBufferSize**, **macPacketOver-**

head listed as default MAC parameters that signify packet size, buffer capacity, and mac frame overhead respectively. As mentioned earlier, the IEEE802.15.6 specifies the physical layer and media access control layer. The MAC has to learn and set its own parameters, and has 12 similar BAN MAC basic parameters and five more parameters that are based on the PHY layer. In Table 12, some are described. The **scheduledAccessLength** defined the slots that a sensor node requests from the hub for planned access. In scheduled access, the **scheduledAccessPeriod** in beacon periods sets how often the scheduled access is requested for.

The parameter **maxPacketTries** is for number of tries for successful reception of packets. For channel to contend **contentionSlotLength** is defined for mini-slots in msecs. To allow more conservative guard times the **enhanceGuardTime** is defined. The parameter **enhanceMoreData** for checking the waiting queue length in buffer, To restrict the polling, the **pollingEnabled** variable can be either enabled or disabled. The **naivePollingScheme** variable is to control polling. When nodes are linked to the coordinator via CSMA, the polling process begins. Upon receiving a polling packet, a node begins transmitting stored data packets.

9 Conclusion

In this paper, a qualitative assessment of MAC protocols for WBAN has been carried out through analyzing the designs of Superframe structure, describing more than one access schemes, and supplying taxonomy based qualitative evaluation of the MAC protocols. Heterogeneous and dynamic traffic loads are used for WBAN applications. TDMA scheduling methods are more suited for emergency data, in which the emergency-based sensor sends an alarm signal to the

Table 12 IEEE 802.15.6: parameters

Parameter	Description
macMaxPacketSize	MAC limit the packet size it can Handle
macBufferSize	The internal MAC buffer size of the packet
macPacketOverhead	Increased overhead (bytes) of network data packets (routes) received by the upper layer
scheduledAccessLength	Slots requested by sensor nodes in the hub to be able to allocate reserved access
scheduledAccessPeriod	Frequency of scheduled access requests (beacon period)
maxPacketTries	The maximum number of packet attempts successfully received
contentionSlotLength	The size of the minislot used in the contention (milliseconds)
enhanceGuardTime	It allows the use of more conservative guard times
enhanceMoreData	How many packets are in the buffer to be processed?
pollingEnabled	The variable that controls whether to try to poll
naivePollingScheme	Variables that control how polling is Performed

body coordinator for slot allocation without conducting contention. Contention-based sensors conduct backoffs to access channels, resulting in greater collision, delay, retransmission of dropped packets, and poorer data dependability. The paper discusses in general WBAN's traffic adaptive MAC protocols. In the form of the traffic efficient approach, the latest IEEE standards for WBANs are addressed. Moreover, classifies the current functional solutions to traffic. Over TMac, Tunable Mac, ZigbeeMAC and BaselineMAC protocols performance parameters are often provided and the strengths and disadvantages of the MAC adaptive protocols are seen over ZigbeeMAC and BaselineMAC protocols. The focus is on Castalia's output measures when different parameters are changed and different models are used.

Declarations

Conflict of interest The author(s) declare(s) that there is no conflict of interest.

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